

MAHIDOL UNIVERSITY

Faculty of Information and Communication Technology



Lecture 4
Digital Transmission

August 2026

ปัญญาของแผ่นดิน Wisdom of the Land

มหาวิทยาลัยมหิดล
Mahidol University

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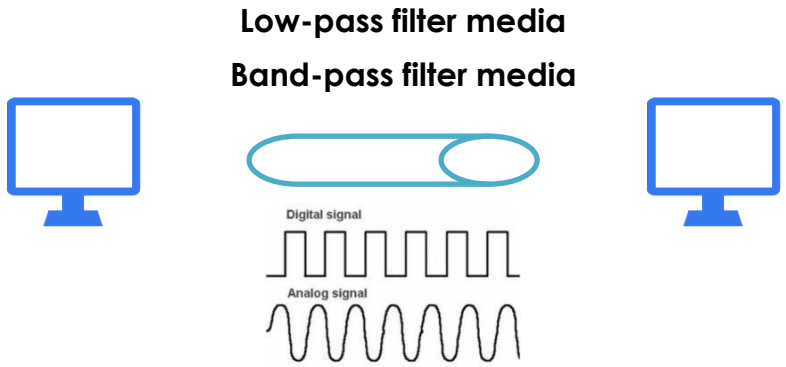
Today Agenda

- Digital-to-Digital Conversion and its issues
  - Baseline wandering
  - DC components
  - Self-synchronization
- Line Coding
  - NRZ, NRZ-I, RZ
  - Manchester Differential Manchester
  - AMI
  - Block Coding (4B/5B)
- Signal Rate vs Data Rate
- Analog-to-Digital Conversion
  - Sampling
  - PAM
  - Quantization
  - PCM



Data to Signal Conversion

- There are two types of media
  - **Low-pass filter media** (e.g., LAN cable, Fiber optics cable)
    - Allow signal of many frequencies to pass.
    - Suitable for digital transmission
  - **Band-pass filter media** (e.g., telephone line, air)
    - Allow signals with specific frequencies to pass.
    - Suitable for analog transmission.

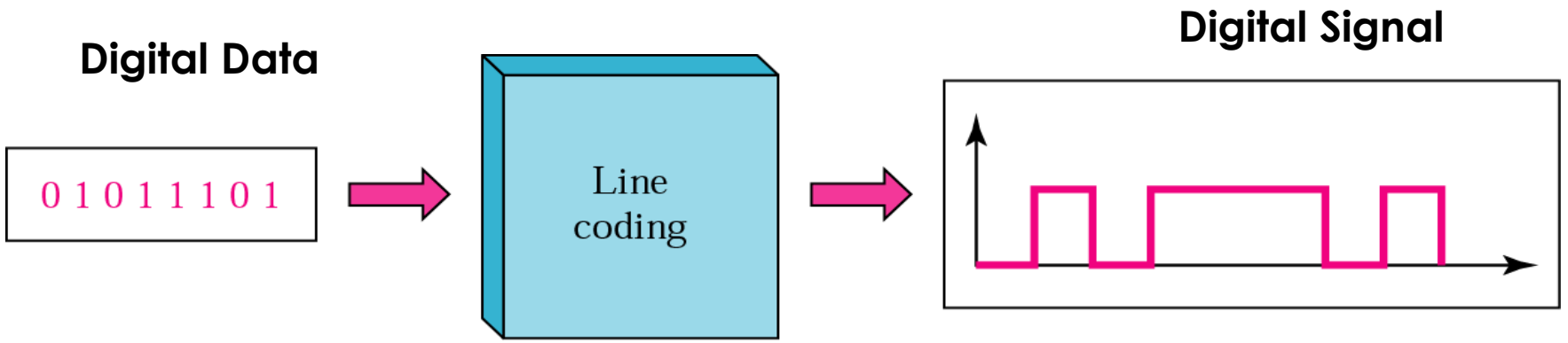


Input \ Output	Digital Signal	Analog Signal
	Digital Transmission Low-pass filter media (support digital signal)	**Analog Transmission** Band-pass filter media (support analog signal)
Digital data	**Digital-to-Digital conversion** (Line coding)	**Digital-to-Analog conversion** (Shift keying)
Analog data	**Analog-to-Digital conversion** (Pulse Code Modulation)	**Analog-to-Analog conversion** (Modulation)

Digital-to-Digital Conversion

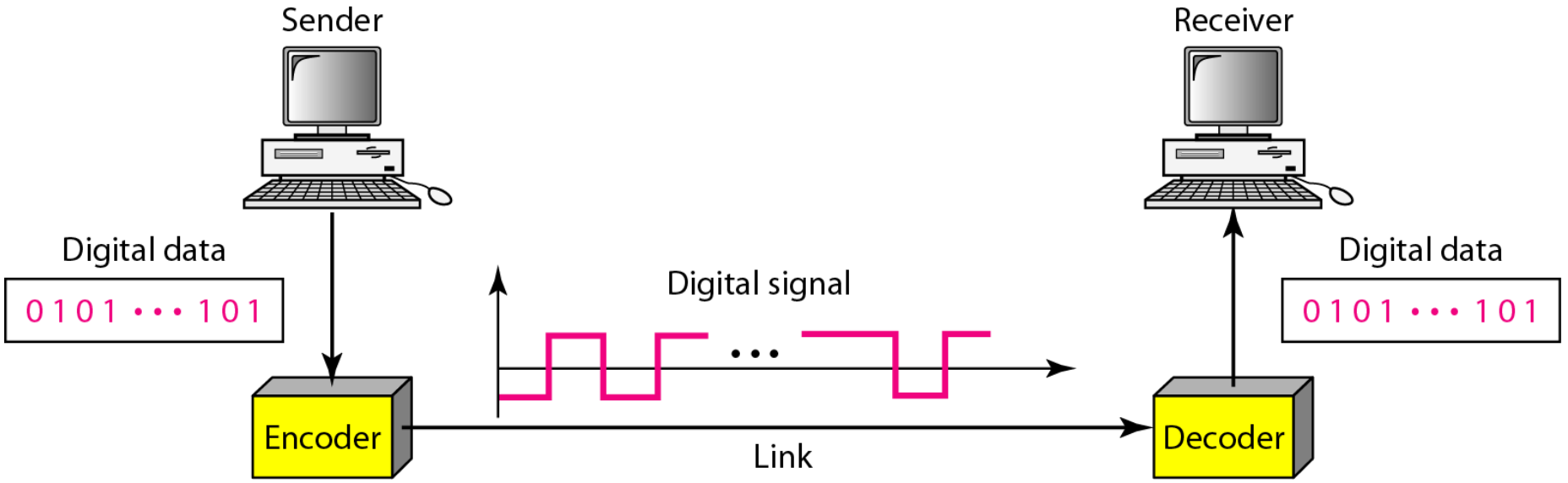
- **When the media can support a wide range of frequency (low-pass filter)**, digital data could be transmitted as digital signals.

- This method is called Line Coding.



Encoding and Decoding

- Encoding: the method of converting original data into a different representation of data.
- Decoding: the method of reversing the conversion of the encoded data to the original data.



Digital-to-Digital Conversion Issues

1. Baseline wandering
2. DC components
3. Self-synchronization



1. Baseline Wandering

- During the decoding of a digital signal, the receiver calculates a running average of the received signal power in order to compare whether the next signal is positive or negative. This average is called baseline.
  - If the baseline is too high, it is difficult to detect positive signal
  - If the baseline is too low, it is difficult to detect negative signal

- A long string of 0s and 1s can cause a drift in the baseline and make it difficult for the receiver to decode correctly. This is called baseline wandering.

- Baseline wandering should be prevented by using good line coding scheme.

Ref: https://www.docsity.com/en/answers/define-baseline-wandering-its-effect-digital-transmission/112549/

1. Baseline Wandering

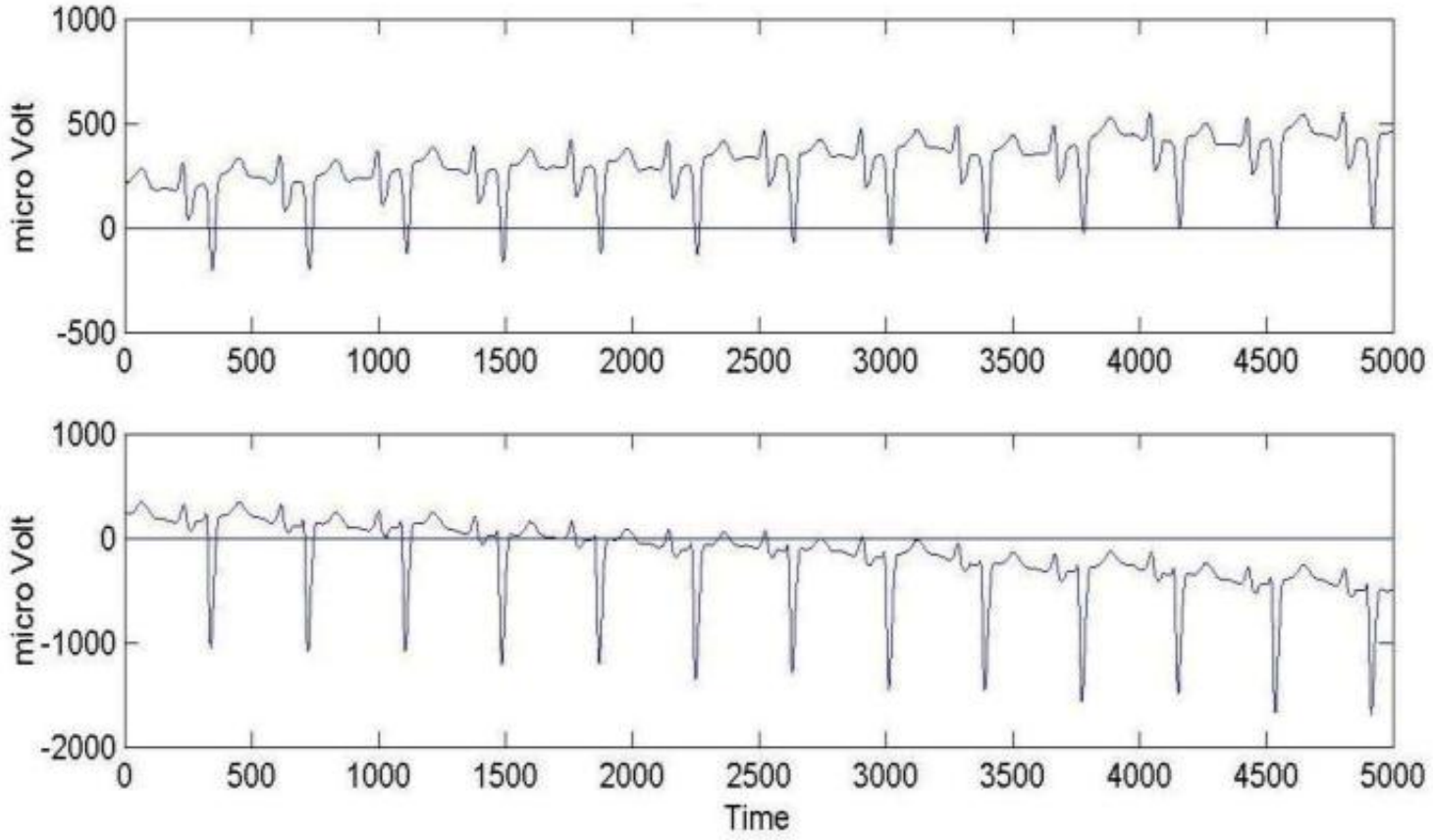


Figure 1. ECG signals with baseline drift [5]

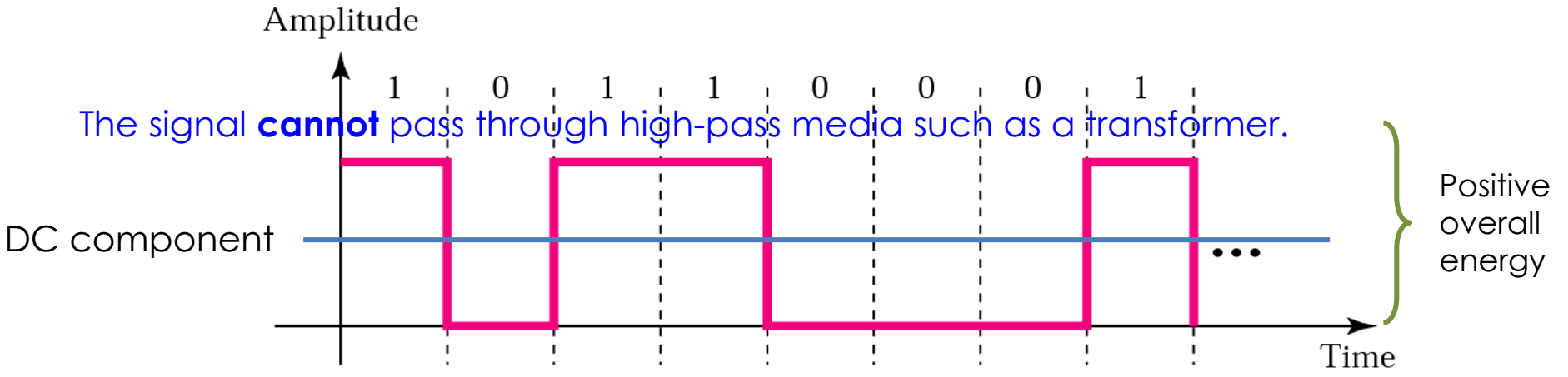
2. DC Components

- When the voltage level in a digital signal is constant for a while, the spectrum creates very low frequencies.

- Some overall systems do have high pass filter, DC components will render the entire signal as low-frequency signals. **Low-frequency signals cannot go through high pass filter channels.**

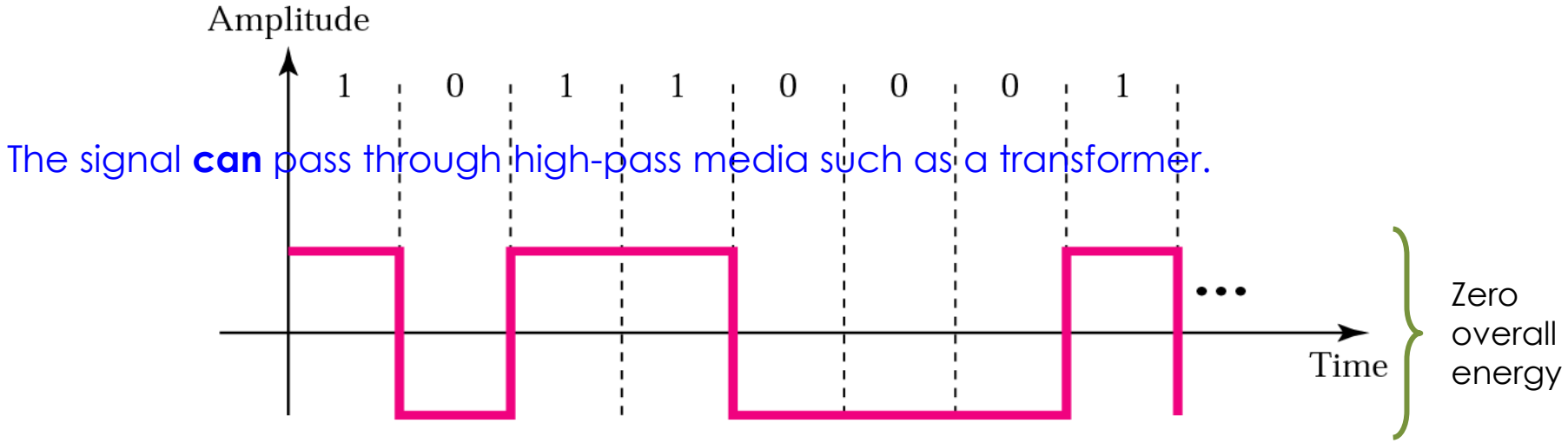
- These frequencies around zero, called DC or direct current components.
- This component is undesirable because
  - If a signal has to pass through a system which does not allow low frequencies to pass, **the output signal is distorted** and may create errors in the data output.
  - This component is extra energy residing on the line and this extra energy is **useless but cause power consumption.**

2. DC Components



The signal **cannot** pass through high-pass media such as a transformer.

a. A signal with dc component

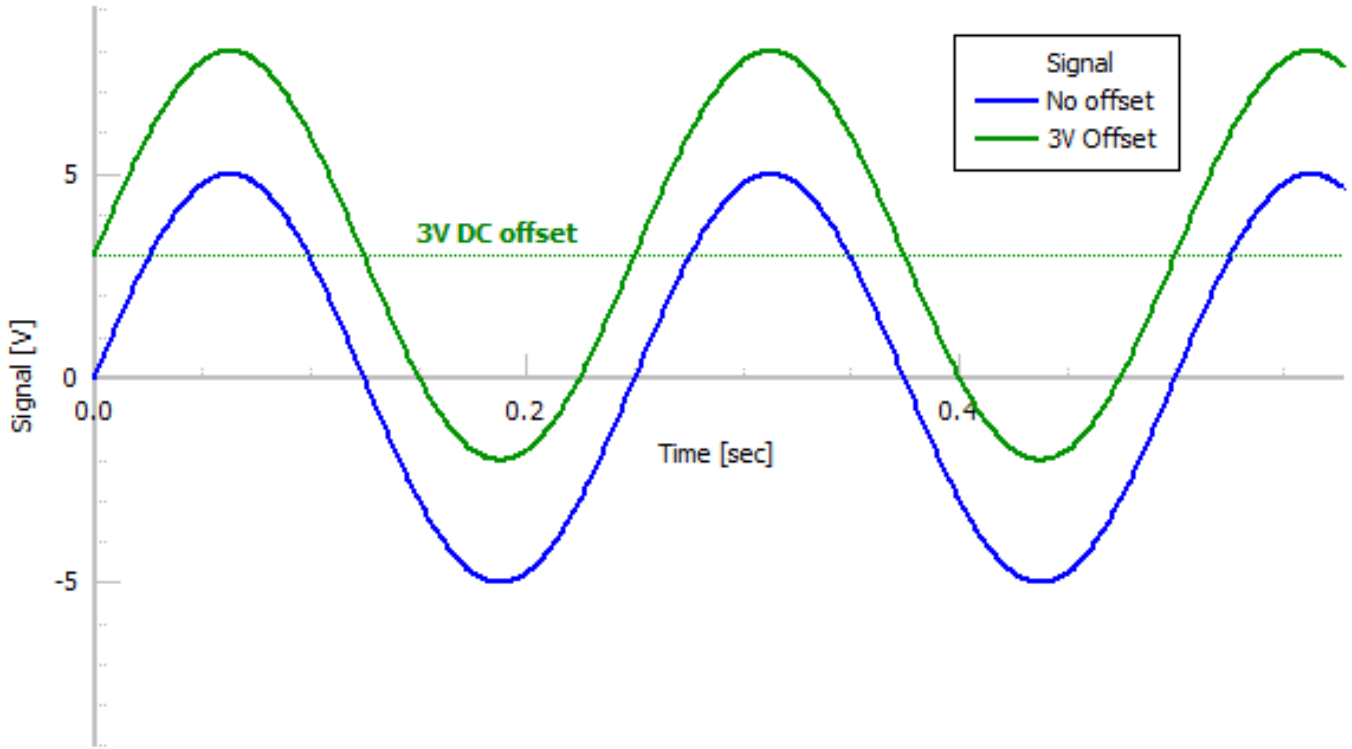


The signal **can** pass through high-pass media such as a transformer.

b. A signal without dc component

2. DC Component

- DC component is in the green signal compare to the blue signal which does not have DC component (fixed voltage)

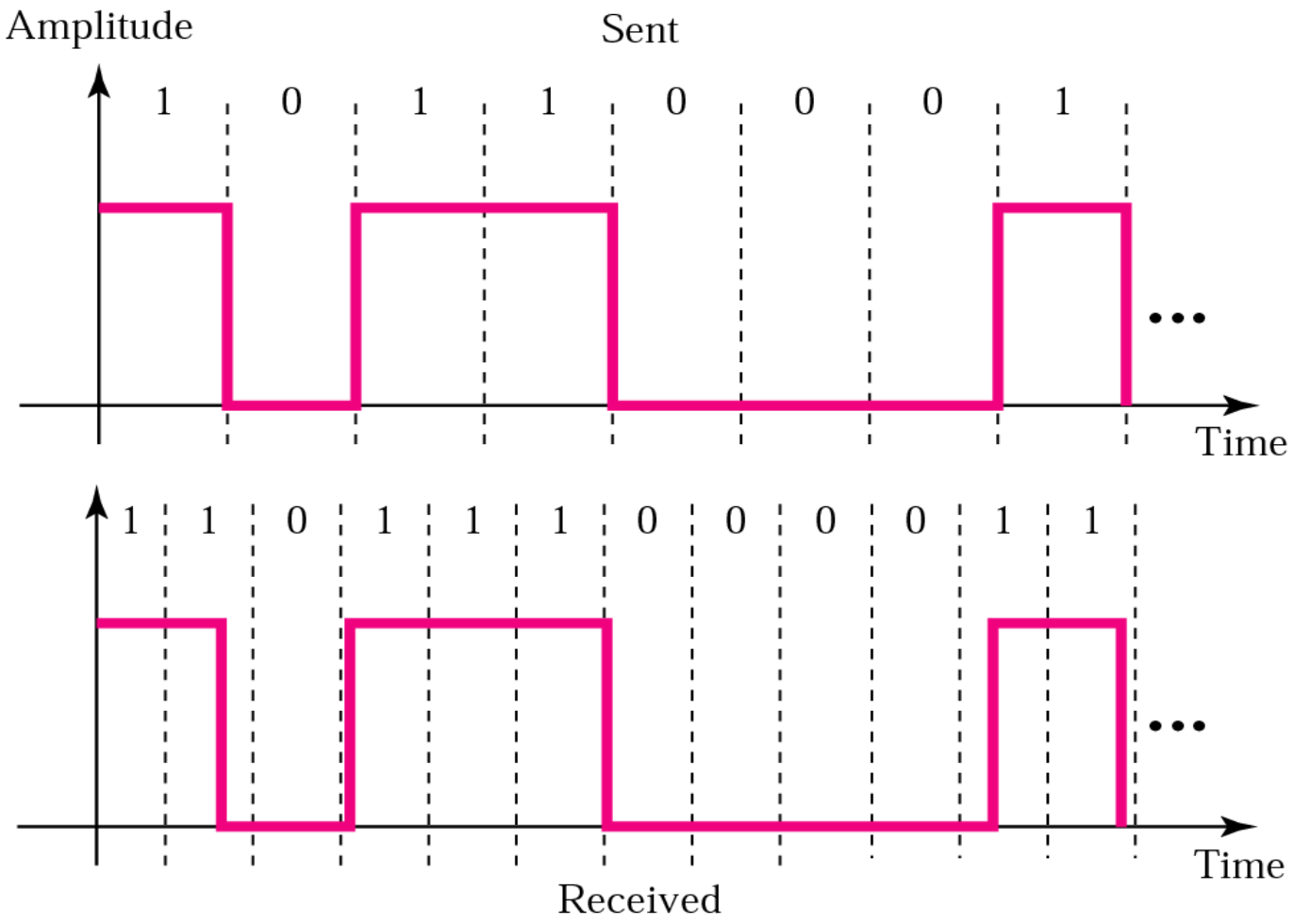


3. Self-Synchronization

- To correctly interpret the signals received from the sender, the receiver's bit intervals must correspond exactly to the sender's bit intervals
- A self-synchronizing digital signal includes **timing information** in the data being transmitted
  - Transition in the signal that alert the receiver to the beginning, middle, or end of the pulse
- If the receiver is faster or slower, the bit intervals are not matched and the receiver might misinterpret the signals.

3. Self-Synchronization

- Lack of synchronization



3. Self-Synchronization

- Example: In a digital transmission, the receiver clock is 0.1 percent faster than the sender clock. How many extra bits per second does the receiver receive if the data rate is 1 Kbps? How many if the data rate is 1 Mbps?

- Solution:
  - At 1 Kbps:
    - 1000 bits sent ➔1001 bits received➔1 extra bps
  - At 1 Mbps:
    - 1,000,000 bits sent ➔1,001,000 bits received➔1000 extra bps

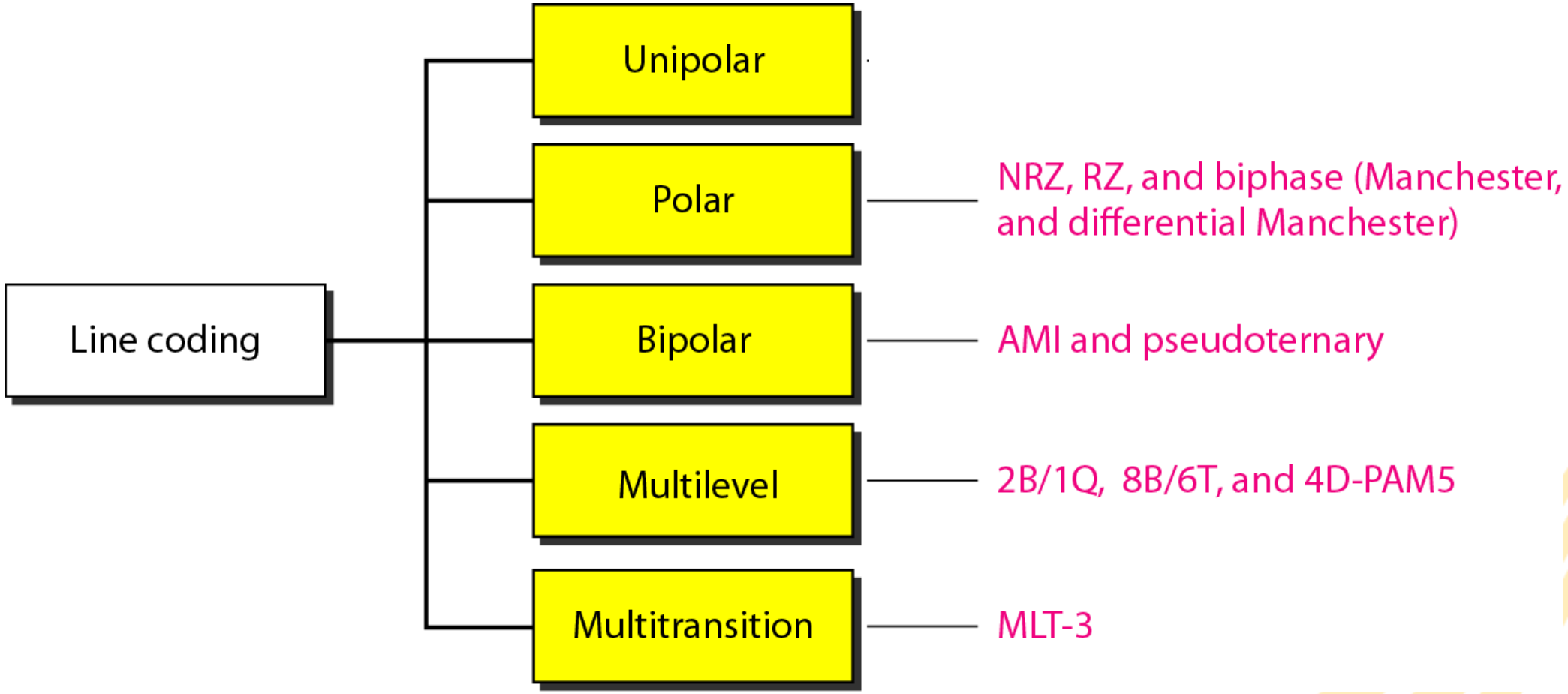
Good Digital Signal Transmission

- No baseline wandering
- No DC component
- Have self-synchronization ability



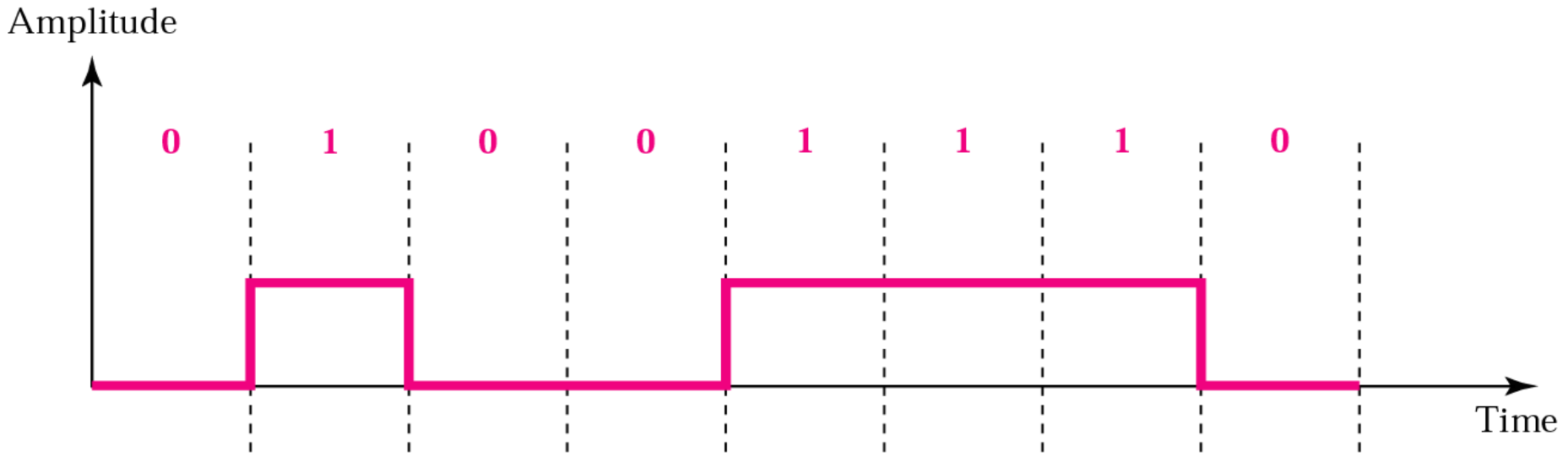
Digital-to-Digital Conversion

- Line coding schemes



Unipolar Encoding

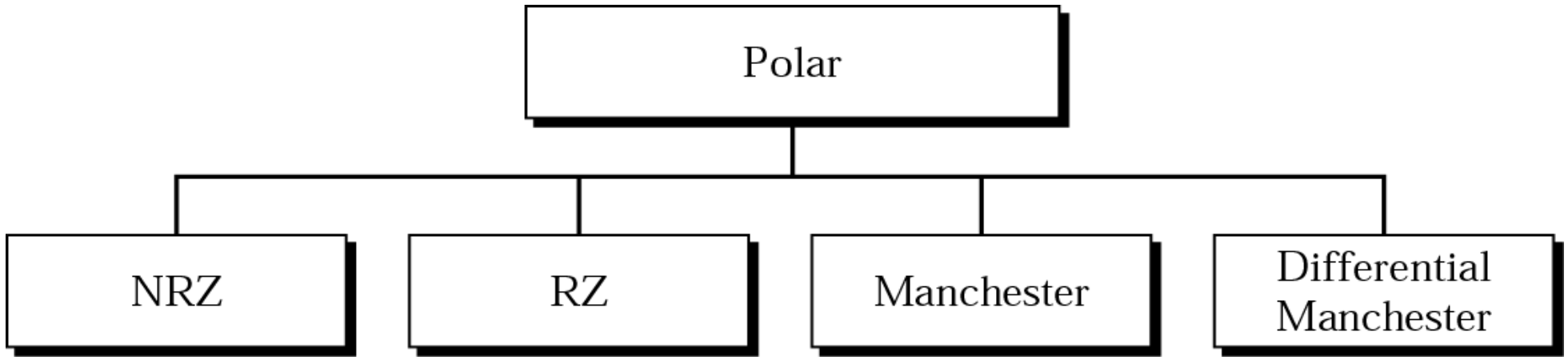
- Unipolar encoding uses only **one voltage level**. The average amplitude of a unipolar encoded signal is nonzero.



- **Issues**:
  - DC Components (poor matching of 0 and 1 voltages)
  - Lack of synchronization (long 0 and 1) causing received data errors.

Polar Encoding

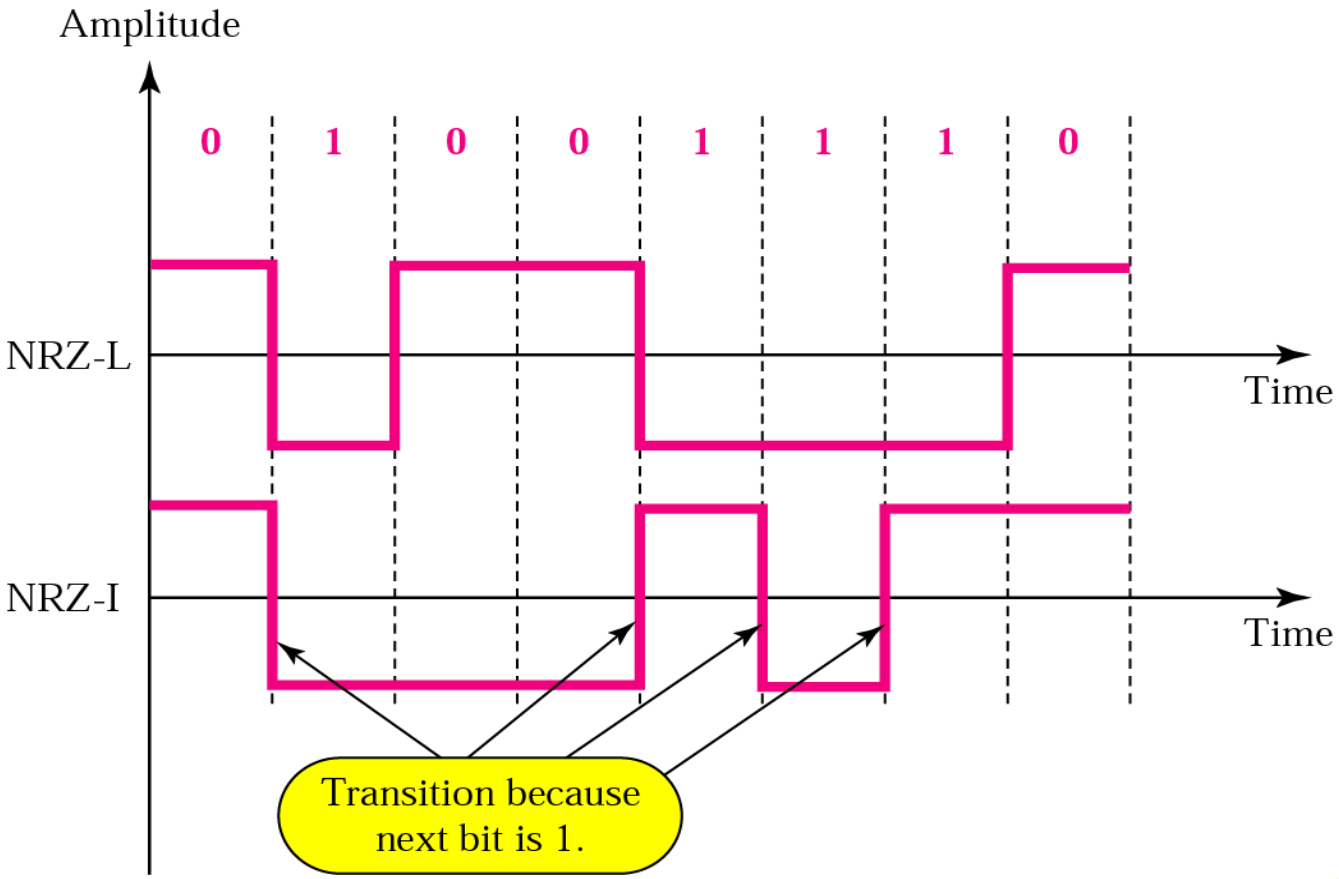
- Polar encoding uses **two voltage levels** (positive and negative).



NRZ (Nonreturn to Zero)

- The signals of this type do not return to zero voltage.
- The value of the signal is always either positive or negative
  - NRZ-L: A long string of 0s, 1s can cause problem with self-synchronization
  - NRZ-I: A long string of 0s can cause problems with self-synchronization

I = Invert



NRZ-L vs. NRZ-I

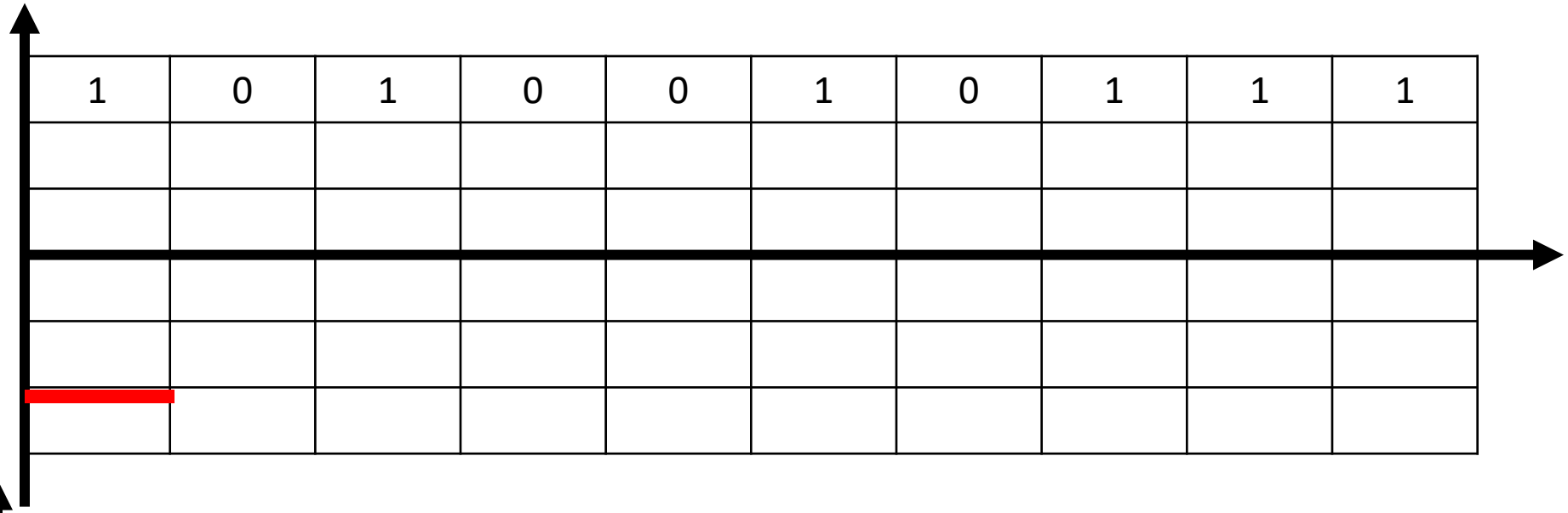
- **NRZ-L (Non Return to Zero – Level)**
  - Represent bit '0' and bit '1' as voltage level.
  - The sender and receiver determines the number of bits sent by counting the number of clock slots.
  - The receiver will detect the same number of bits sent by the sender as long as the clock of the receiver is the same as the clock of the sender (which is impossible in the real world). There is synchronization problem.

- **NRZ-I (Non Return to Zero – Invert)**
  - Partially solve a problem of NRZ-L by changing the bit-1 symbol meaning (semantics) from the voltage level to the inversion the voltage level representing bit 1.
  - NRZ-I is superior to NRZ-L due to the synchronization property provided by the signal change each time a bit '1' is encountered

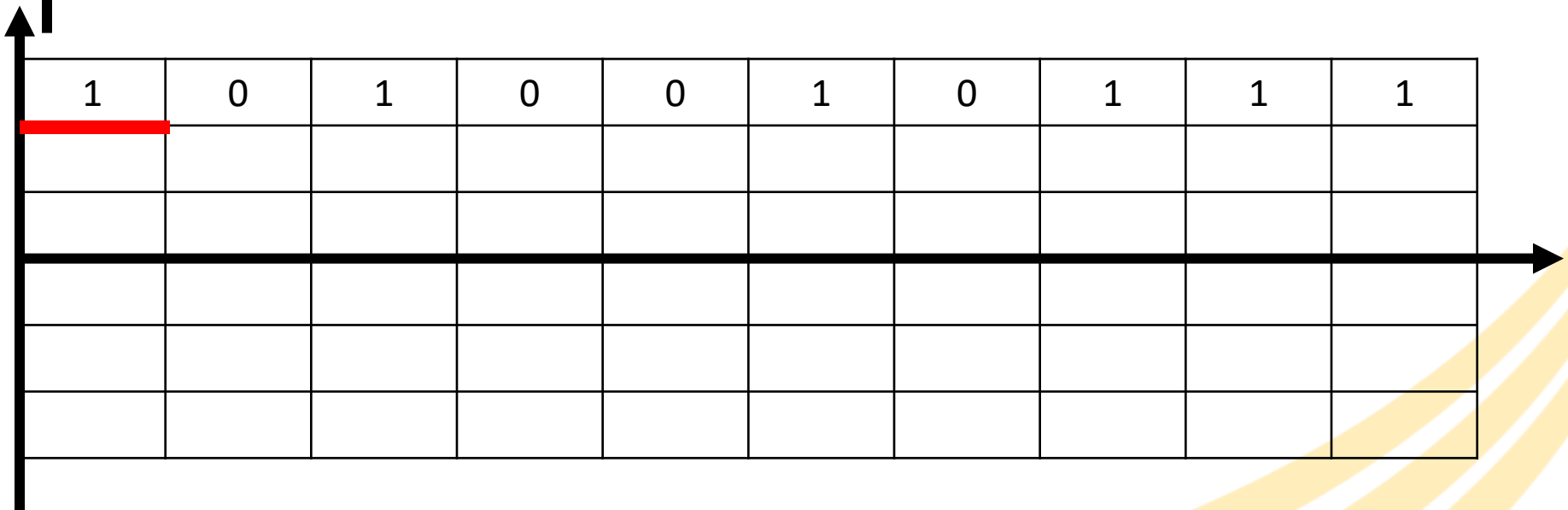
NRZ-L and NRZ-I Exercise 1

- Draw the signal graph when the sender what to send data "**1010010111**" by using NRZ-L and NRZ-I.

NRZ-L



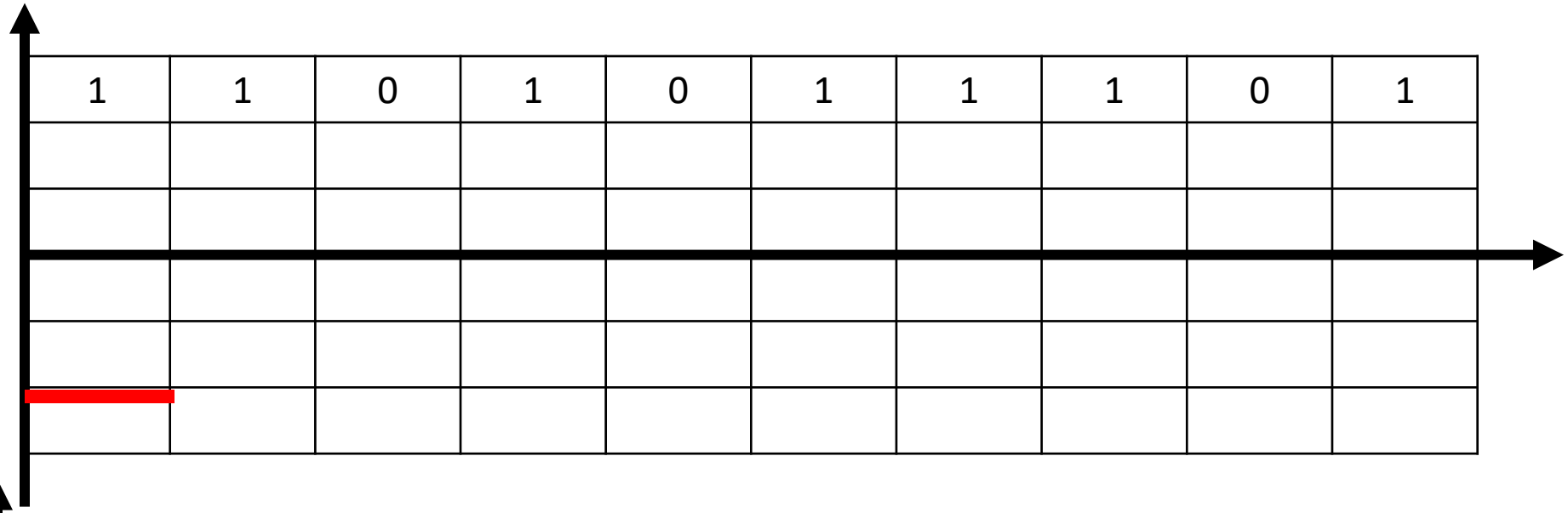
NRZ-I



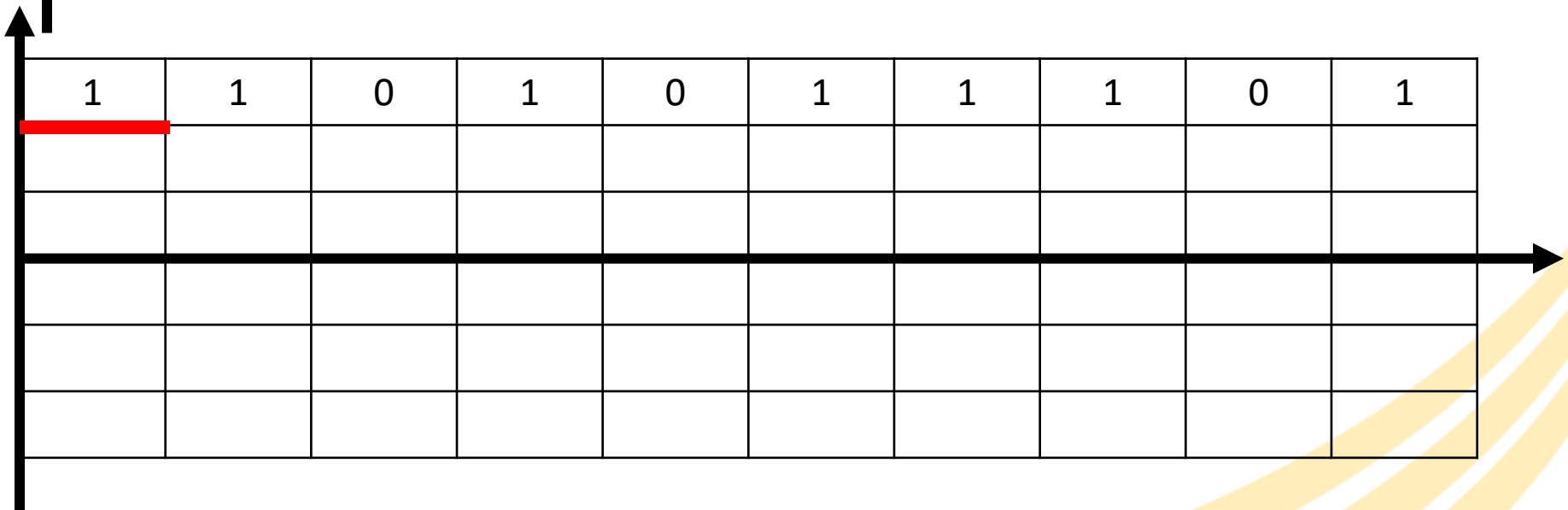
NRZ-L and NRZ-I Exercise 2

- Draw the signal graph when the sender what to send data by using NRZ-L and NRZ-I.

NRZ-L



NRZ-I



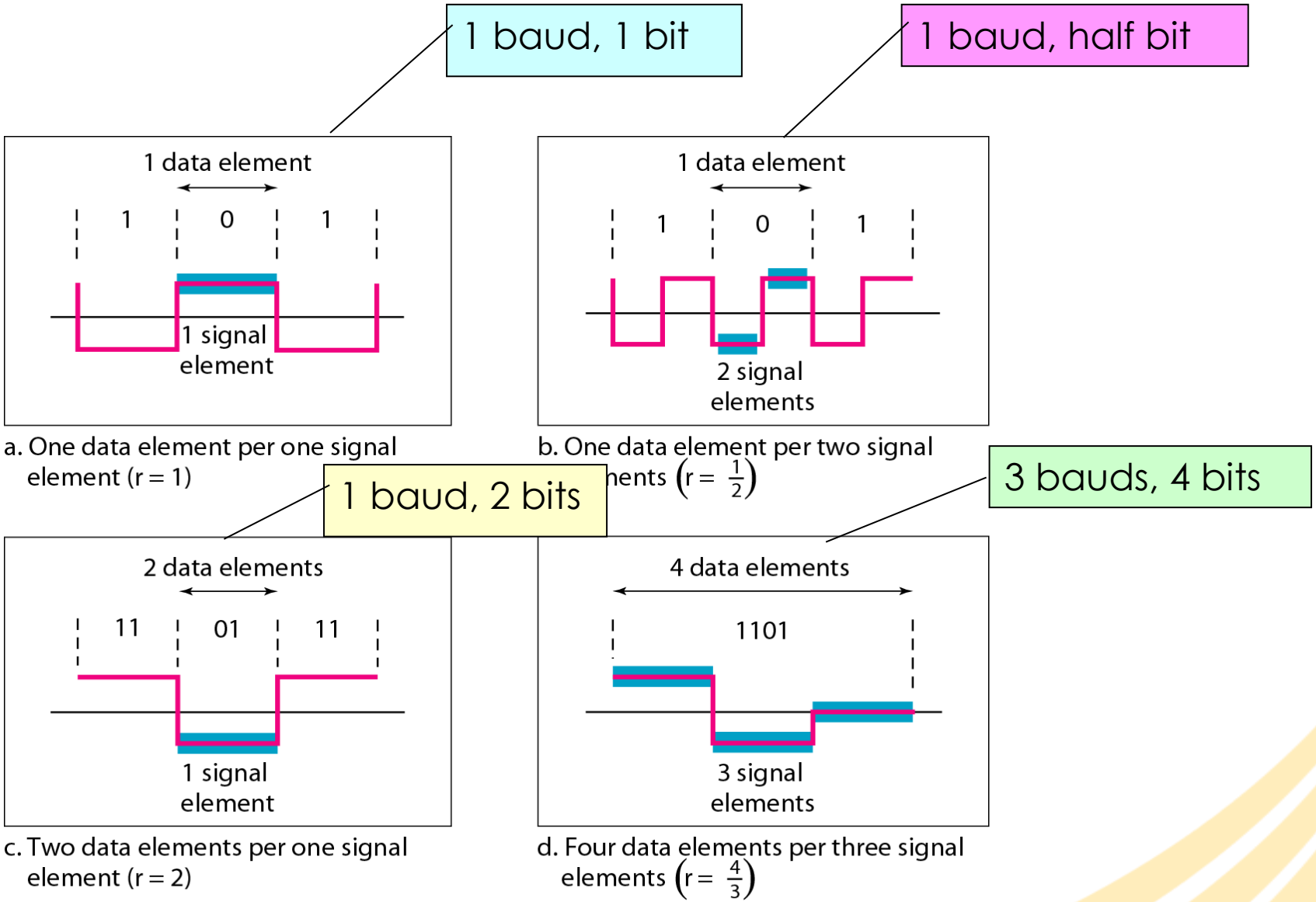
Signal Rate vs. Data Rate

- Data rate is the number of data elements (bits) sent in 1 second. It is also called bit rate.

- Signal rate is the number of signal elements (symbols) sent in 1 second. It is also called baud rate or baud or pulse rate or modulation rate.



Signal Rate vs. Data Rate



Signal Rate vs. Data Rate

- According to the relationship between signal rate and data rate below, our goal is to increase the data rate while decreasing the signal rate in order to minimize the energy usage and actual signal transmission time.

$$S_{avg} = c \times N \times \frac{1}{r}$$

- S = number of signal element (signal rate or baud rate; baud/s)
- c = case factor (0 == best; 1 == worst; average = 0.5)
- N = data rate = bit rate (bps)
- r = bits per baud or bits per signal

Example

- A signal is carrying data in which one data element is encoded as one signal element (r = 1). If the bit rate is 100 kbps, what is the average value of the baud rate if c is 1/2?

- Solution: $S = c \times N \times \frac{1}{r} =$

Exercise

- A signal is carrying data in which one data element is encoded as two signal element (r = 1/2). If the bit rate is 200 kbps, what is the average value of the baud rate if c is 1/2?

- Solution: $S = c \times N \times \frac{1}{r} =$

Bandwidth Requirements

- Although the actual bandwidth of a digital signal is infinite, the effective bandwidth is finite.

$$\mathrm{B_{min}} = S = c \times N \times \frac{1}{r}$$

$$\mathrm{N_{max}} = \frac{1}{c} \times \mathrm{B} \times r$$

- B_{min} = minimum bandwidth = baud rate
- N_{max} = maximum data rate
- c = case factor
- r = bits per baud or bits per symbol

Example

- The maximum data rate of a channel is $N_{max} = 2 \times B \times \log_2 L$ (defined by the Nyquist formula). Does this agree with the previous formula for N_{max}?

- Solution:
  - A signal with L levels actually can carry $\log_2 L$ bits per level ($r = \log_2 L$). If each level corresponds to one signal element and we assume the average case ($c = 1/2$), then we have

$$\mathrm{N_{max}} = \frac{1}{c} \times \mathrm{B} \times r = 2 \times \mathrm{B} \times \log_2 L$$

$$\text{Bit rate} = 2 \times \text{Baud rate} \times \log_2 L$$

Example 1

- Two signals (A and B) have two and four data levels with a pulse duration of 1 ms. The pulse rates and bit rates of the signals are as follows

- Signal A : **2** data levels

$$\text{pulse rate} = B = \frac{1}{1\text{ ms}} = \frac{1}{10^{-3}} = 1{,}000\text{ baud/s}$$

$$\text{bit rate} = N = 2 \times B \times \log_2 L = 2 \times 1{,}000 \times \log_2 2 = 2{,}000\text{ bps}$$

- Signal B : **4** data levels

$$\text{pulse rate} = B = \frac{1}{1\text{ ms}} = \frac{1}{10^{-3}} = 1{,}000\text{ baud/s}$$

$$\text{bit rate} = N = 2 \times B \times \log_2 L = 2 \times 1{,}000 \times \log_2 4 = 4{,}000\text{ bps}$$

Example 2

- Signal A has 16 data levels with a pulse duration of 5 ms. The pulse rates and bit rates of the signals are as follows

- Signal A : 16 data levels

$$\text{pulse rate} = B = \frac{1}{5\ \text{ms}} = \qquad \text{baud}$$

$$\text{bit rate} = N = 2 \times B \times \log_2 L = 2 \times _____ \times \log_2 16 = _____ \ \text{bps}$$

Example 3

- A system is using NRZ-I to transfer 10-Mbps data. What are the average signal rate and minimum bandwidth?

- Recall

$$\mathrm{B_{min}} = S = c \times N \times \frac{1}{r}$$

- We have

$$\mathrm{B_{min}} = S = \frac{1}{2} \times 10{,}000{,}000 \times \frac{1}{\log_2 2} = \boxed{} \text{ baud/s}$$

- We can conclude that the minimum bandwidth is $\boxed{}$ MHz.

Let’s do the exercise and have a Break !





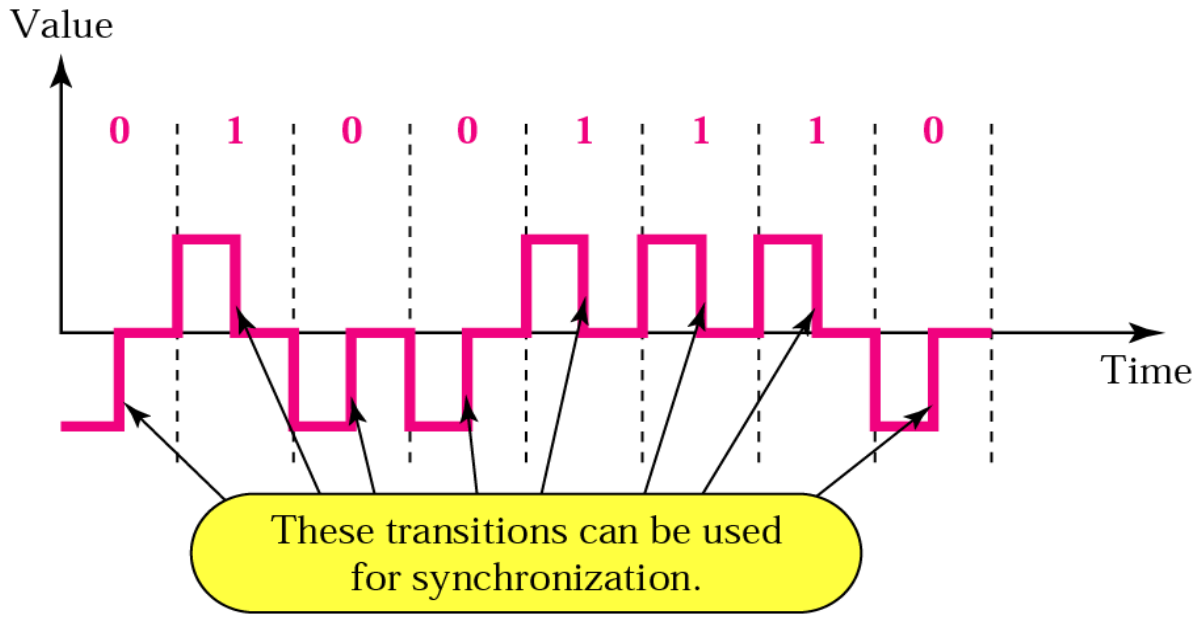






Return to Zero (RZ)

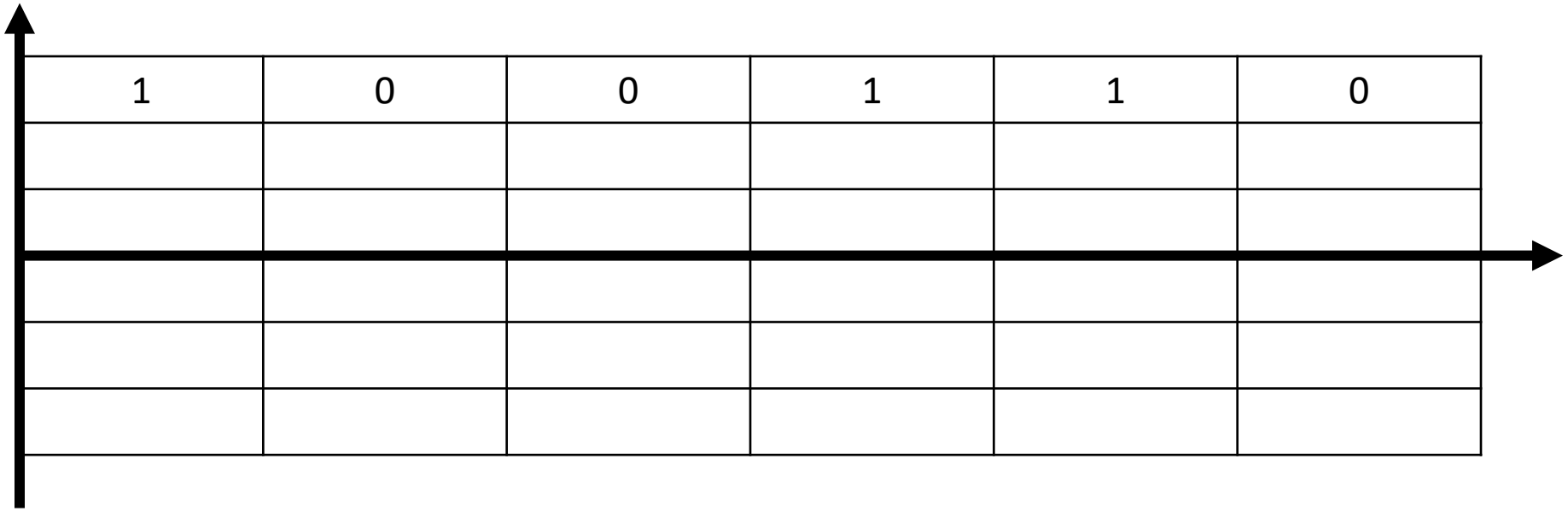
- RZ uses three voltage values: positive, negative and zero.
- For each symbol, bit detection is based on voltage change (similar to NRZ-I).
  - The voltage change is between the symbol representing a bit.
  - The voltage change always **return to zero**.
- Requires two signal changes to encode 1 bit, therefore occupies more bandwidth



RZ Encoding Exercise 1

- Draw the signal graph when the sender what to send data "**100110**" by using RZ encoding.

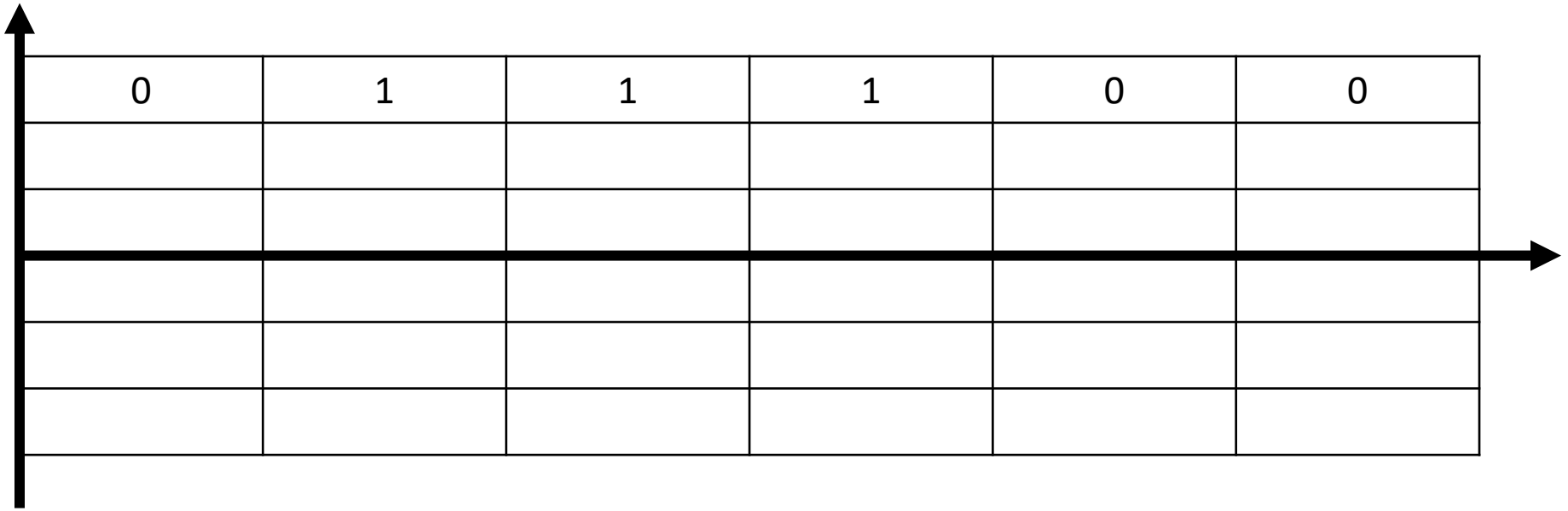
RZ



RZ Encoding Exercise 2

- Draw the signal graph when the sender what to send data by using RZ encoding.

RZ



Manchester Encoding

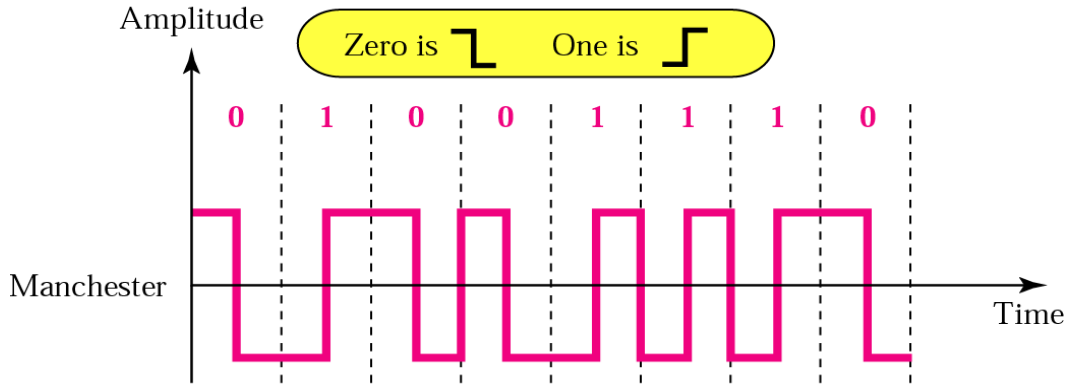
- Manchester Encoding uses an inversion at the middle of each bit interval for both synchronization and bit representation
  - 1: negative-to-positive
  - 0: positive-to-negative

- By using a single transition for a dual purpose, Manchester encoding achieves the same level of synchronization as RZ but with only two levels of voltage.

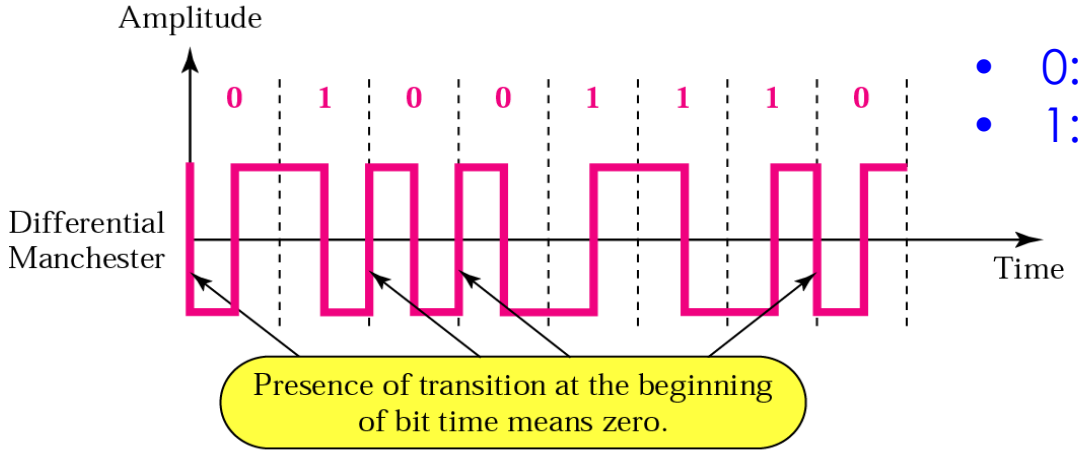
- There are two types of Manchester encoding
  - Manchester encoding
  - Differential Manchester encoding

Manchester Encoding

- In **regular Manchester encoding**, the transition at the middle of the bit is used for both synchronization and bit representation.



- In **differential Manchester encoding**, the transition at the middle of the bit is used only for synchronization. The bit representation is defined by the inversion or noninversion at the beginning of the bit

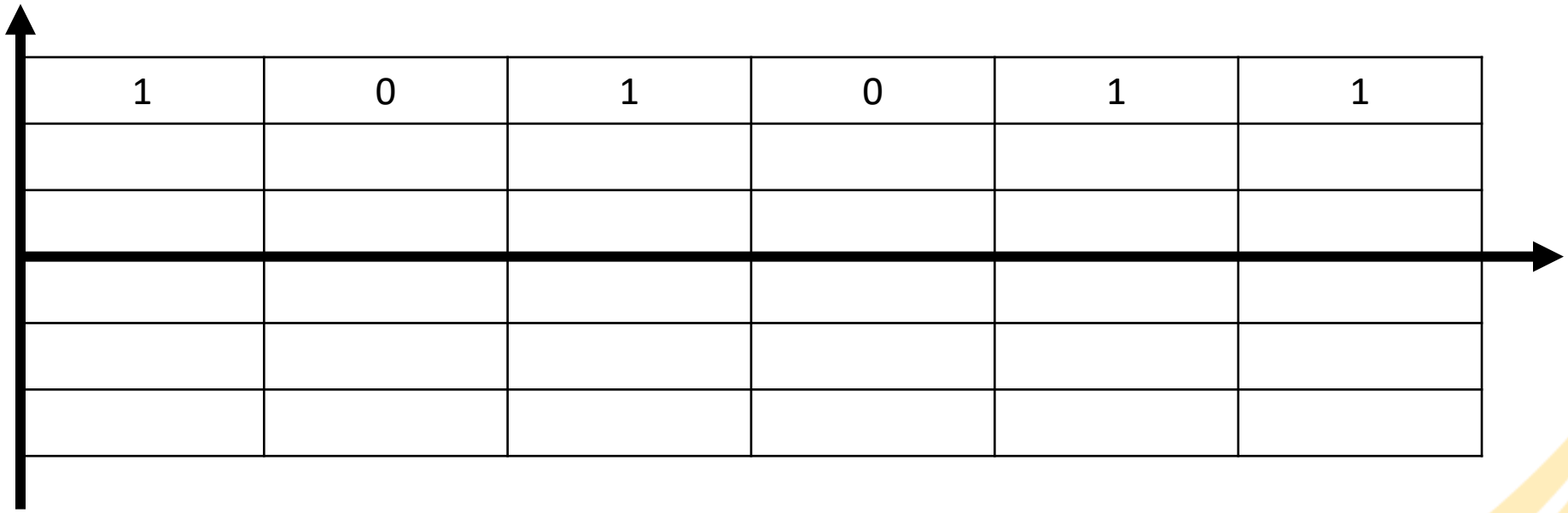


- 0: a transition
- 1: no transition

Differential Manchester Exercise 1

- Draw the signal graph when the sender what to send data "**101011**" by using the Differential Manchester encoding
- Note: We usually start drawing the signal (left most voltage) with high voltage. (see previous)

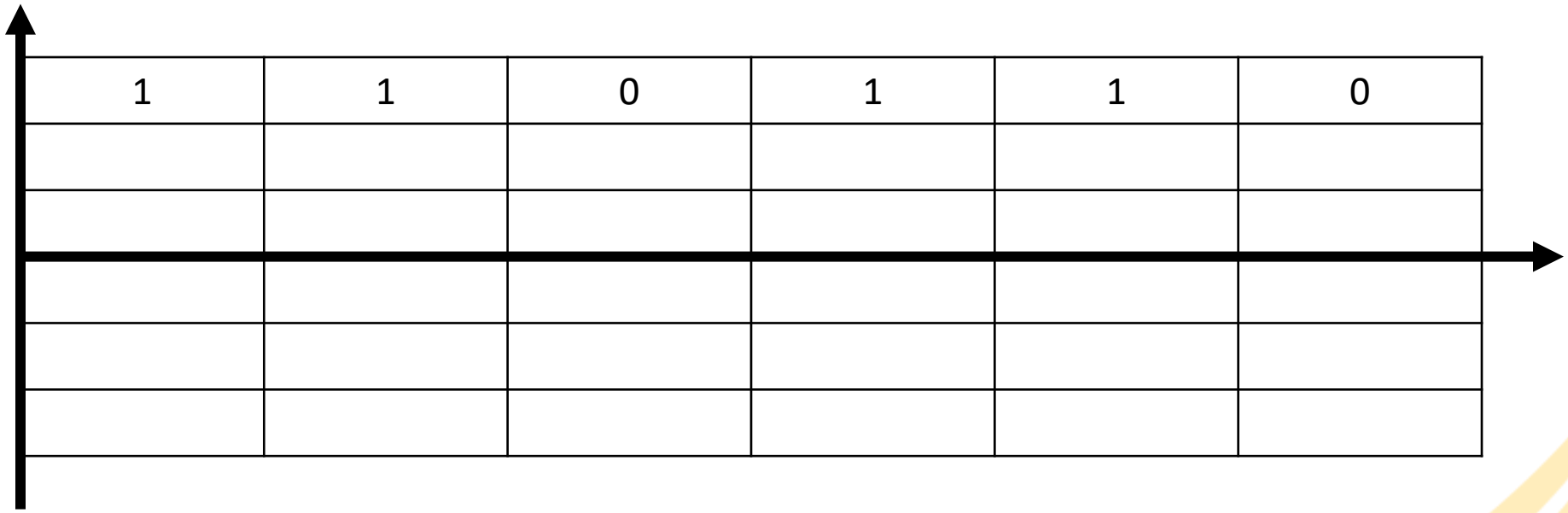
Differential Manchester



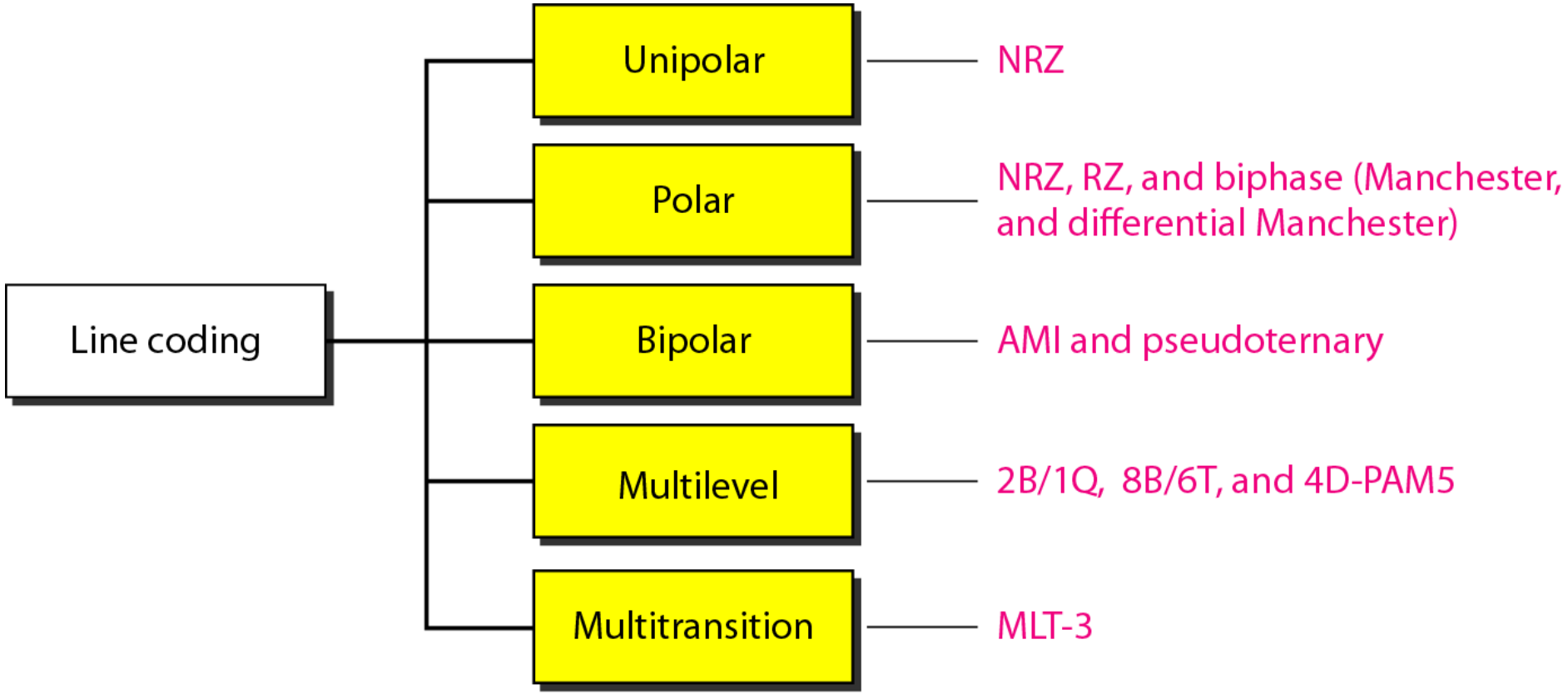
Differential Manchester Exercise 2

- Draw the signal graph when the sender what to send data by using the Differential Manchester encoding
- Note: We usually start drawing the signal (left most voltage) with high voltage. (see previous)

Differential Manchester

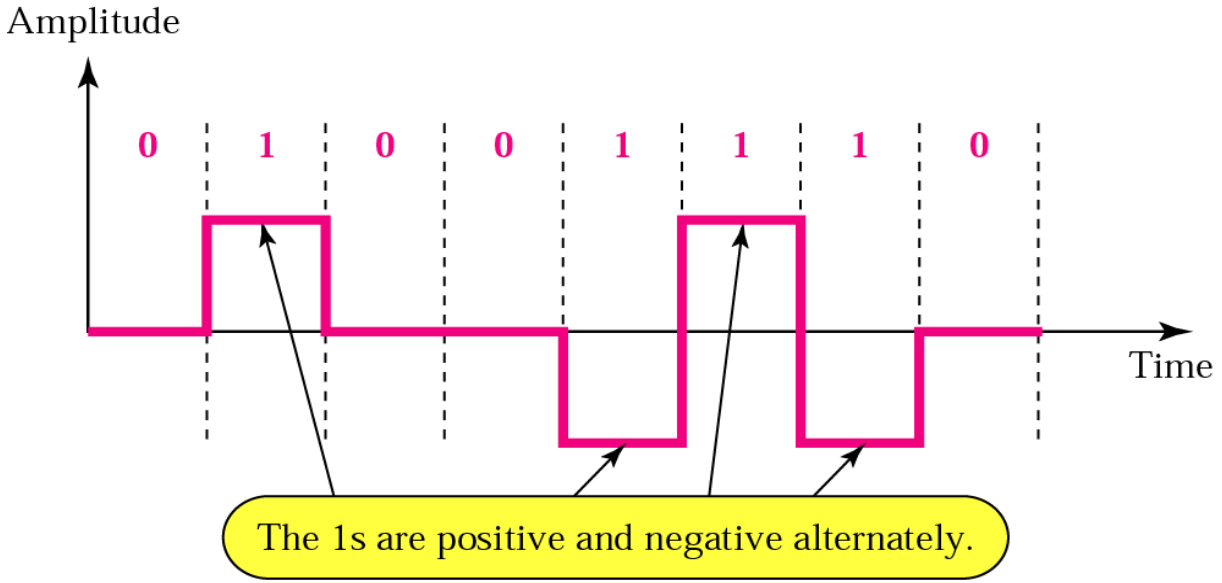


Line Coding Schemes



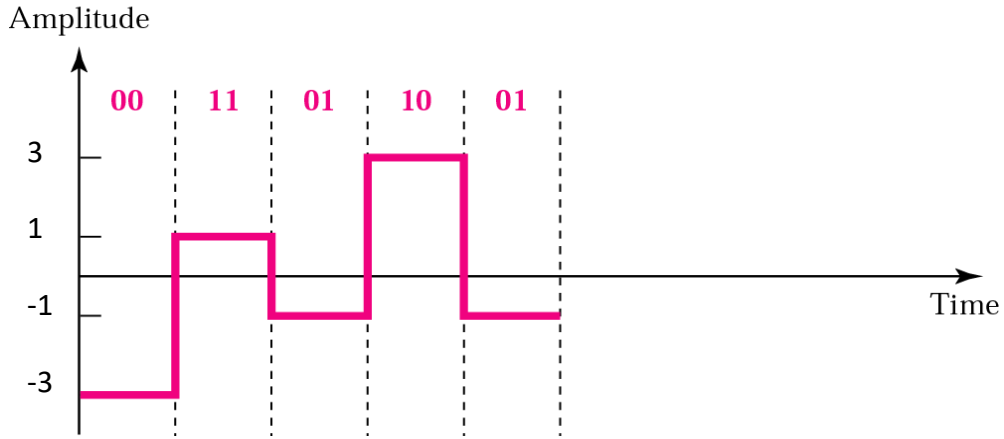
Bipolar AMI Encoding

- Bipolar AMI encoding (**Alternate Mask Inversion**) uses three levels: positive, zero, and negative.
  - A binary 0 is encoded as zero volts as in unipolar encoding.
  - A binary 1 is encoded alternately as a positive voltage and a negative voltage. The alternating positive and negative voltage for bit 1 of bipolar AMI prevents the build-up of DC component since the positive and negative voltage cancel each other out in the long run.
- Unlike RZ, the zero level in bipolar encoding is used to represented binary 0.



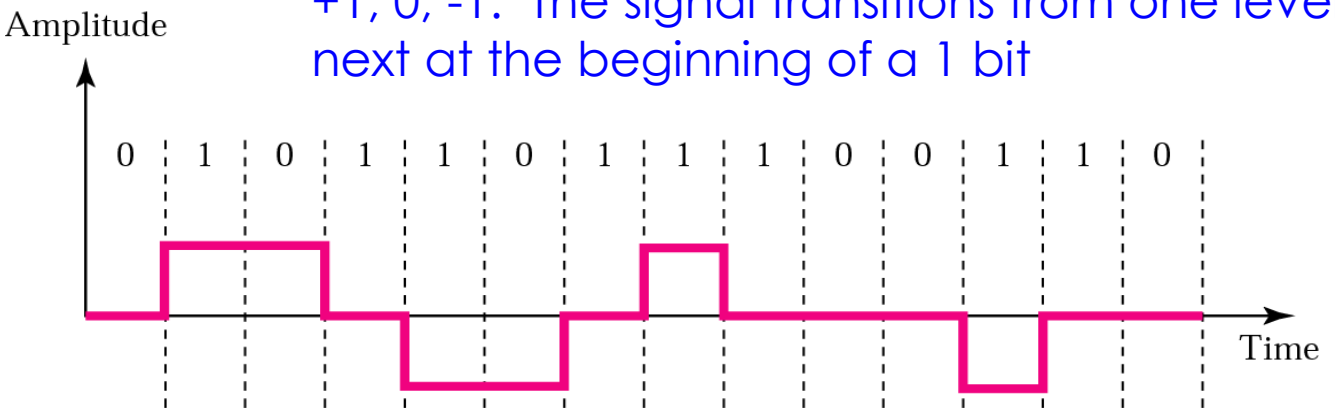
Multilevel 2B1Q and Multitransition MLT-3

- **2B1Q**: 2 bits data, 1 4-voltage-level symbol (quaternary)



- **MLT-3**: Multiline transmission using 3 levels

+1, 0, -1: The signal transitions from one level to the next at the beginning of a 1 bit



Block Coding

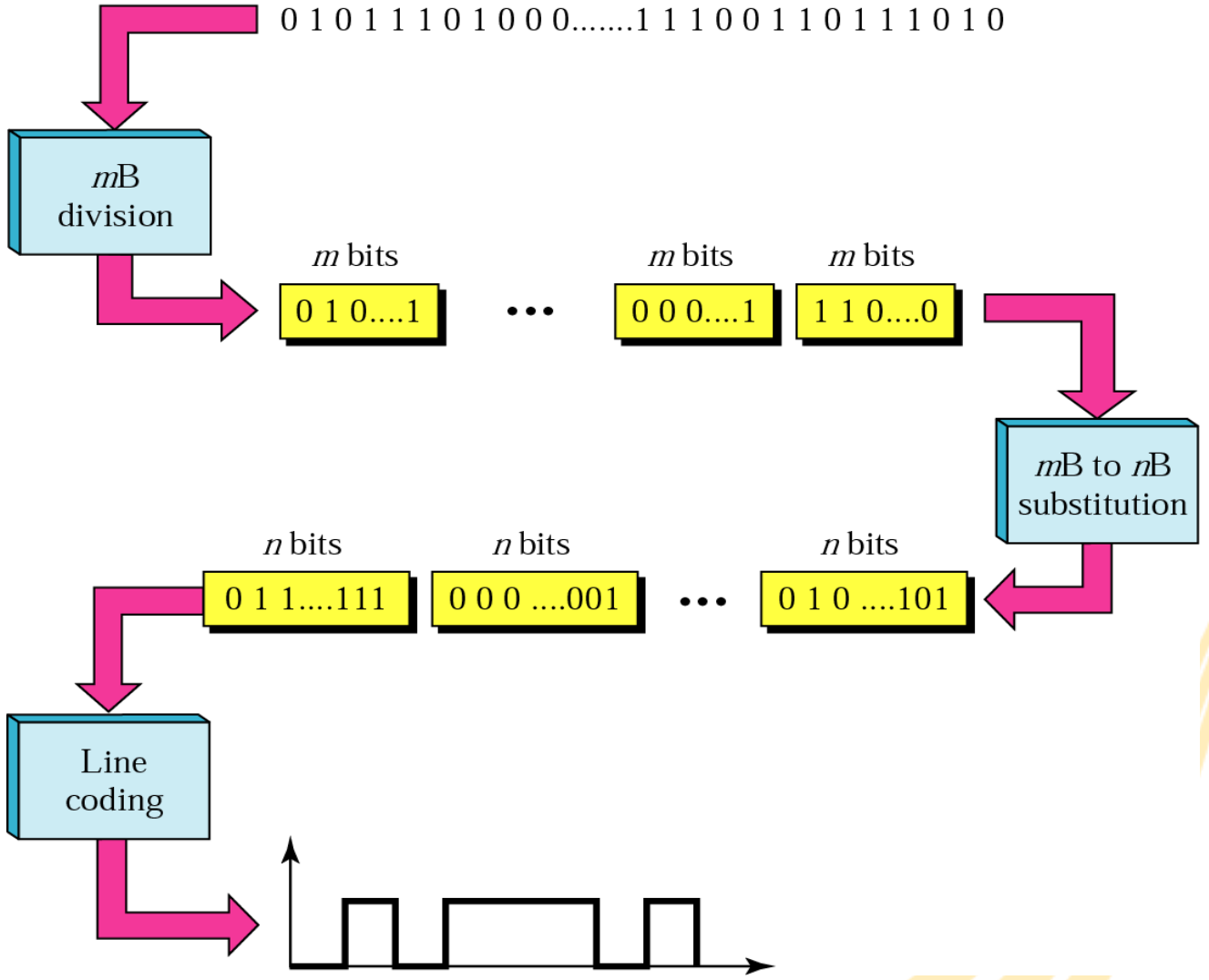
- **Data Pre-processing (Data manipulation)**: In order to prevent the problem of long consecutive zeroes or ones, data bits that are going to be sent may have to be converted into the form that does not have long consecutive zeroes or ones.

- This conversion may **add extra bits** to the original data but the benefits of reducing error outweigh the drawback of redundancy overhead.

- To improve performance of line coding, we need some kind of redundancy to ensure synchronization

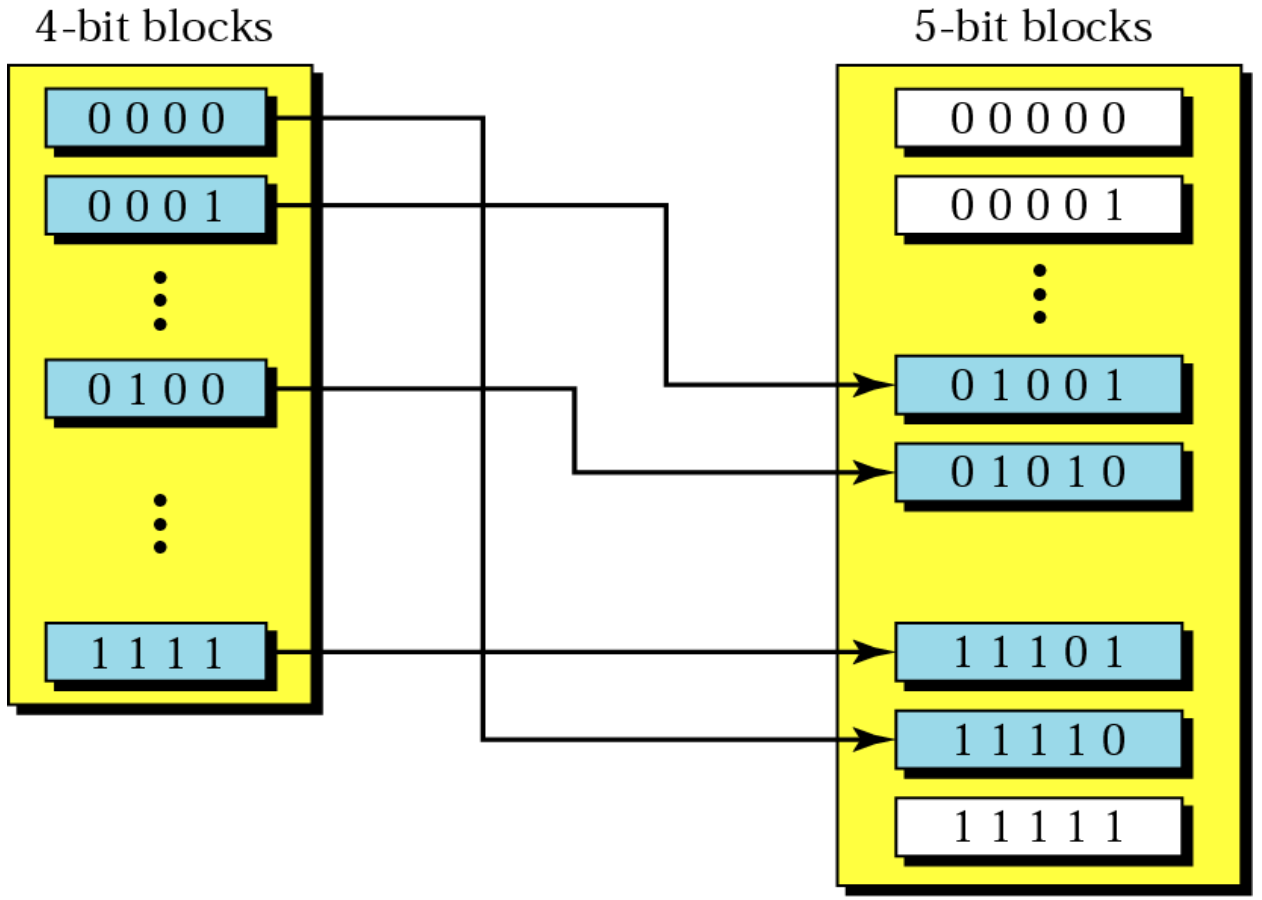
Block Coding

- 3 Steps
  1. Division of data into m-bit blocks
  2. Substitution of m-bit blocks into n-bit blocks using block mapping table
  3. Converting data into signal using regular line coding techniques



Substitution in Block Coding

- In this example, 4-bit blocks (2^4 = 16 possible patterns) are replaced by 5-bit blocks (2^5=32 patterns).



Common Block Code Mapping

- **4B/5B**
  - Replacing 4 bits by 5 bits pattern. Every 4 bits of data is encoded into 5-bit code.
  - Not all 5-bit patterns were used, especially the patterns that contain 3 consecutive 0s.
  - No more than 3 consecutive 0s are encountered.
  - Normally line coded using NRZ-I

- **8B/10B**
  - Replacing 8 bits by 10 bits pattern. Every 8 bits of data is encoded into 10-bit code.

4B/5B Encoding Table

Encoding of data block

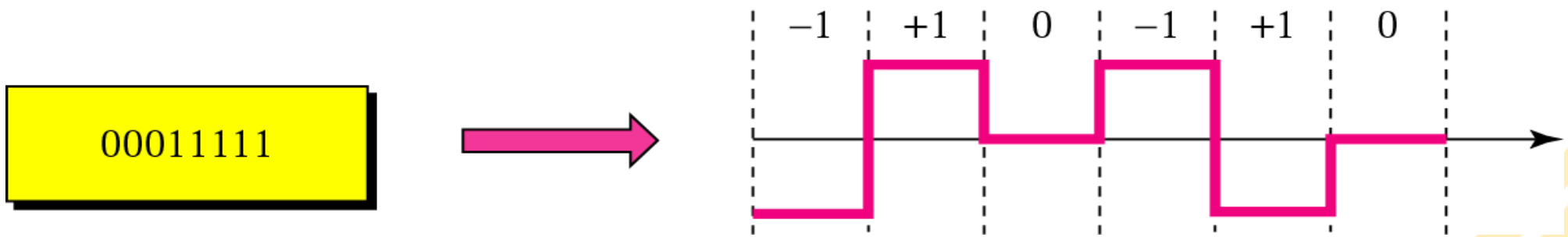
Data	Code	Data	Code
0000	11110	1000	10010
0001	01001	1001	10011
0010	10100	1010	10110
0011	10101	1011	10111
0100	01010	1100	11010
0101	01011	1101	11011
0110	01110	1110	11100
0111	01111	1111	11101

4B/5B Encoding Table

Data	Code
Q (Quiet)	00000
I (Idle)	11111
H (Halt)	00100
J (start delimiter)	11000
K (start delimiter)	10001
T (end delimiter)	01101
S (Set)	11001
R (Reset)	00111

8B/6T Encoding

- 4B/5B and 8B/10B requires more symbol rate in order to transmit data (roughly 25% increase)
- We can improve the transmission performance by increasing the number of bits that a symbol can represent.
- In this case, we substitute 8-bit group with a 6-symbol code which can use 3 different voltage level (positive, negative and zero).
- Thus, we are mapping $2^8 = 256$ patterns to $3^6 = 729$ patterns. The rules of having no consecutive zeros and synchronization support remain.



8B/6T Code Table (Partial)

Data Octet	6T Code Group	Data Octet	6T Code Group	Data Octet	6T Code Group	Data Octet	6T Code Group
00	+-00+-	10	+0+--0	20	00-++-	30	+-00-+
01	0+-+-0	11	++0-0-	21	--+00+	31	0+--+0
02	+-0+-0	12	+00-0-	22	++-0+-	32	+-0-+0
03	-0++-0	13	0++-0-	23	++-0-+	33	-0+-+0
04	-0+0+-	14	0++--0	24	00+0-+	34	-0+0-+
05	0+--0+	15	++00--	25	00+0+-	35	0+-+0-
06	+-0-0+	16	+0+0--	26	00-00+	36	+-0+0-
07	-0+-0+	17	0++0--	27	--+++-	37	-0++0-
08	-+00+-	18	0+-0+-	28	-0-++0	38	-+00-+
09	0-++-0	19	0+-0-+	29	--0+0+	39	0-+-+0
0A	-+0+-0	1A	0+-++-	2A	-0-+0+	3A	-+0-+0
0B	+0-+-0	1B	0+-00+	2B	0--+0+	3B	+0--+0
0C	+0-0+-	1C	0-+00+	2C	0--++0	3C	+0-0-+
0D	0-+-0+	1D	0-+++-	2D	--00++	3D	0-++0-
0E	-+0-0+	1E	0-+0-+	2E	-0-0++	3E	-+0+0-
0F	+0--0+	1F	0-+0+-	2F	0--0++	3F	+0-+0-

4B/5B + NRZ-I Exercise

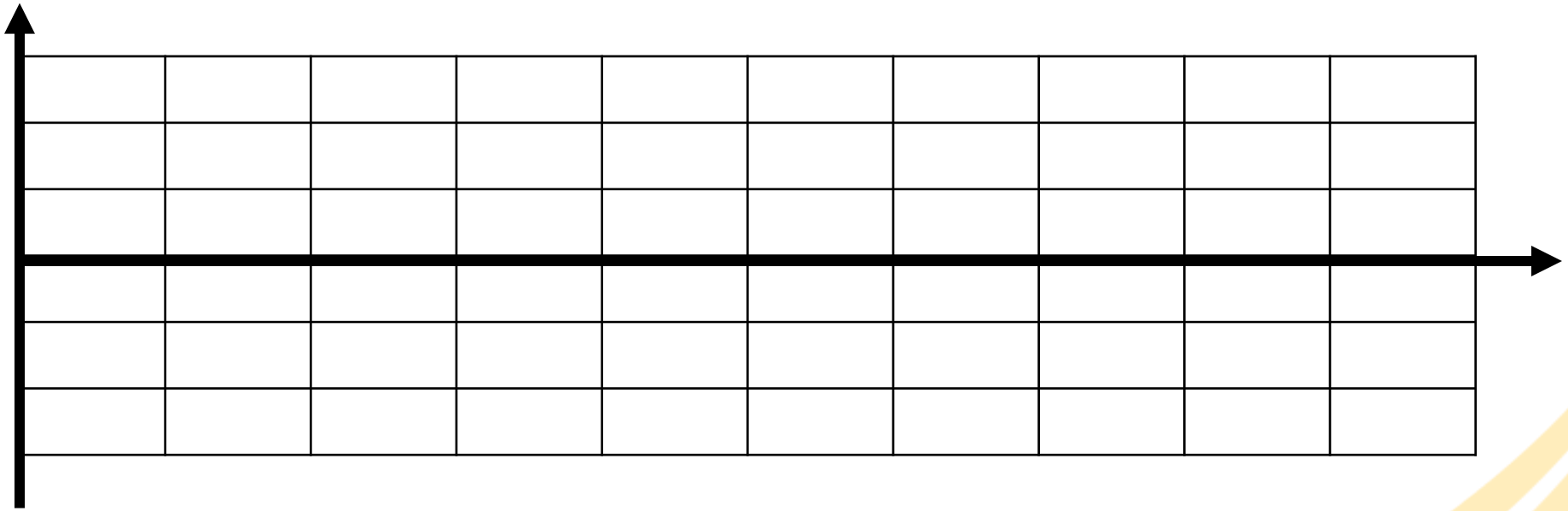
- Use the 4B/5B to encode the data "**01001011**", and use the NRZ-I for line coding.

Data	Code	Data	Code
0000	11110	1000	10010
0001	01001	1001	10011
0010	10100	1010	10110
0011	10101	1011	10111
0100	01010	1100	11010
0101	01011	1101	11011
0110	01110	1110	11100
0111	01111	1111	11101

4B/5B 01001011 → _____ _____

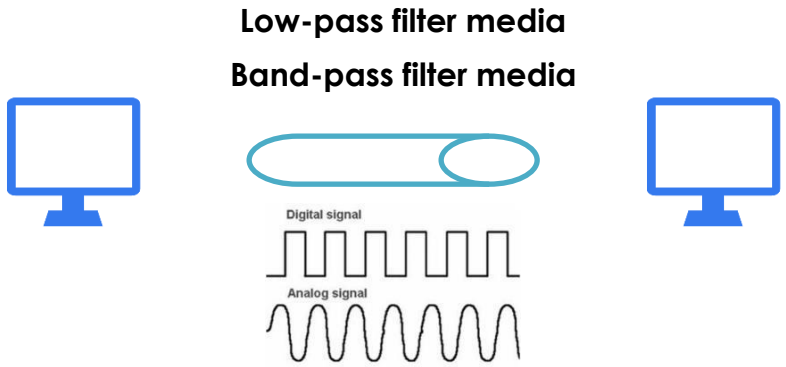
Assume: start with positive voltage
1: transition 0: no transition

NRZ-I



Data to Signal Conversion

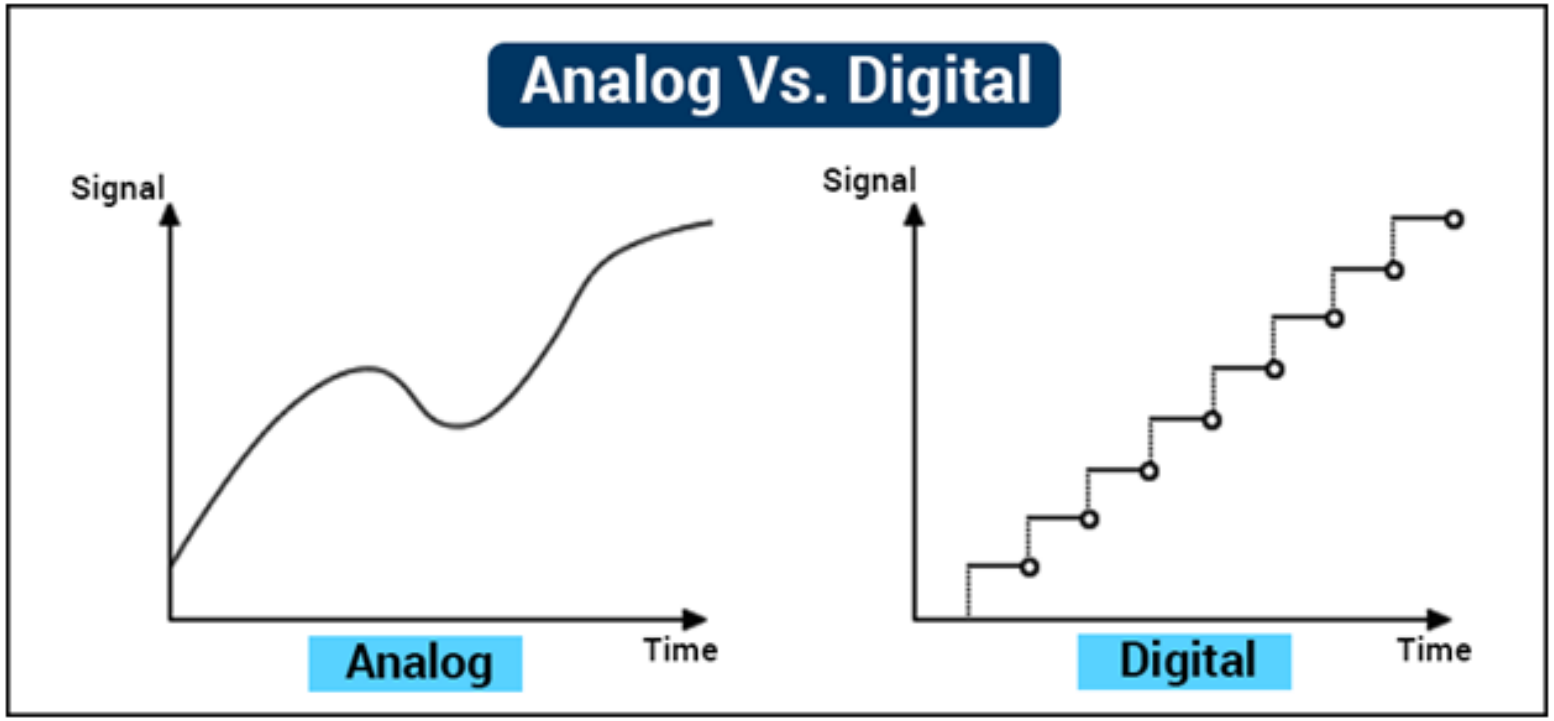
- There are two types of media
  - **Low-pass filter media** (e.g., LAN cable, Fiber optics cable)
    - Allow signal of many frequencies to pass.
    - Suitable for digital transmission
  - **Band-pass filter media** (e.g., telephone line, air)
    - Allow signals with specific frequencies to pass.
    - Suitable for analog transmission.



Output / Input	Digital Signal	Analog Signal
	Digital Transmission Low-pass filter media (support digital signal)	**Analog Transmission** Band-pass filter media (support analog signal)
Digital data	**Digital-to-Digital conversion** (Line coding)	**Digital-to-Analog conversion** (Shift keying)
Analog data	**Analog-to-Digital conversion** (Pulse Code Modulation)	**Analog-to-Analog conversion** (Modulation)

Analog Signals

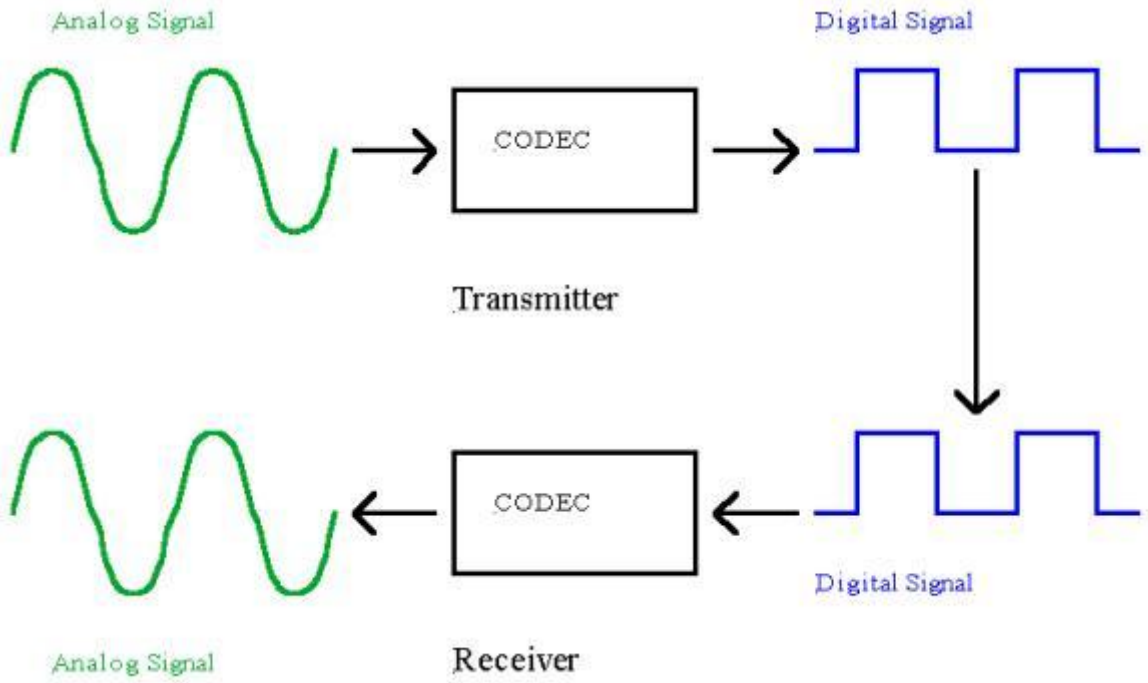
- Analog signals are signals that can take any value (continuous)



Analog-to-Digital Conversion

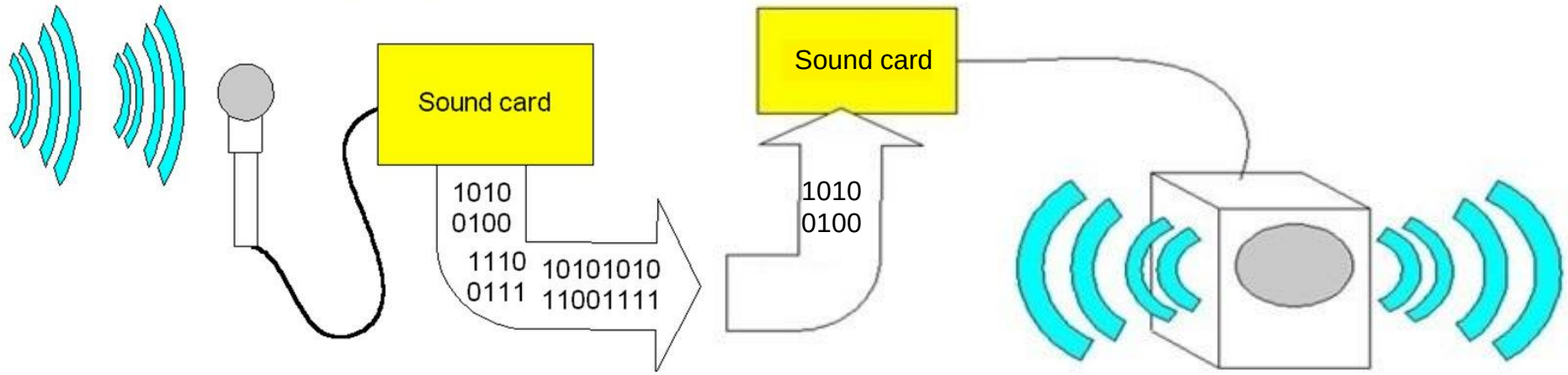
- Digital transmission has many advantages such as
  - Noise tolerance
  - Easier to read incoming signal

- We try to send signals in the form of digital signals using digital channels (low-pass channel)



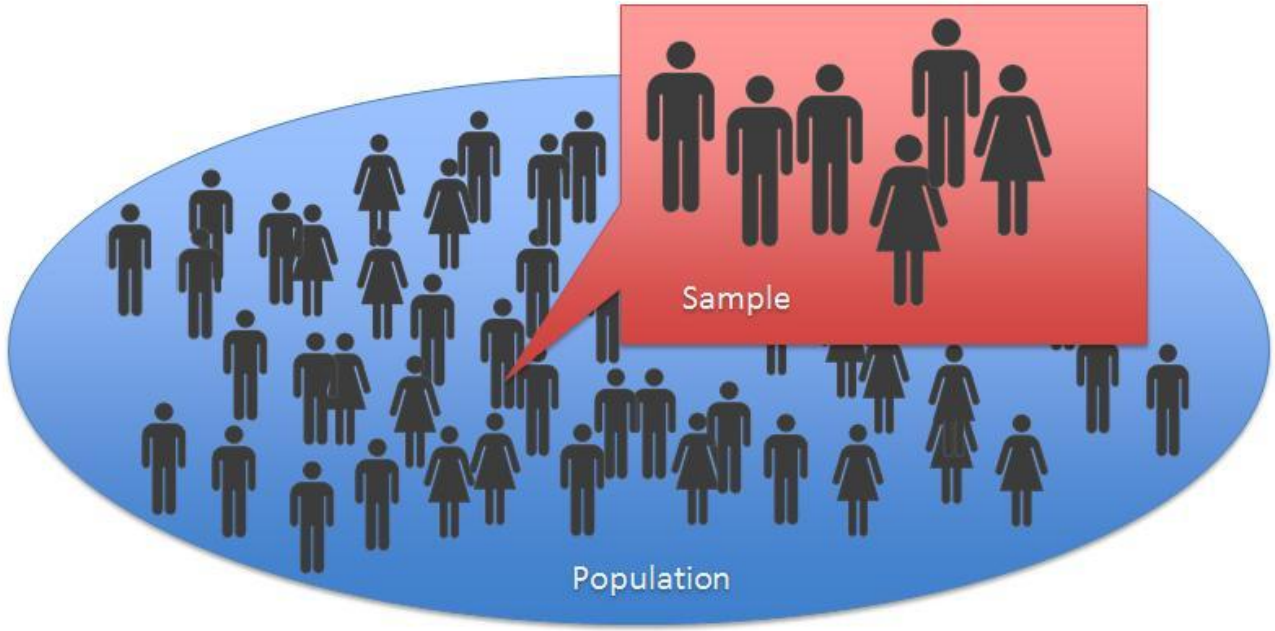
Analog-to-Digital Conversion

- Analog data such as voice is digitized into bits of information which is sent as digital signals.
- The receiver receives digital signal, converts the signal into stream of bits and sent to sound card to convert digital data into analog signals (voice)



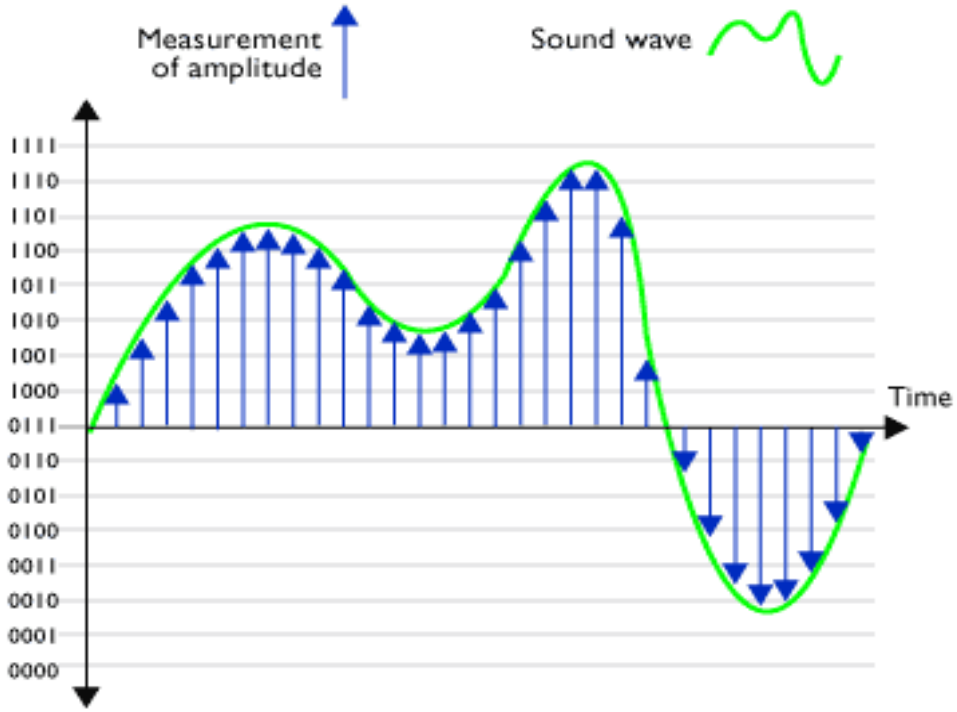
Sampling

- Sampling is the process of selecting a small number of elements from population.
- The selected elements are called samples.
- The group of samples should be a good representative of the population.



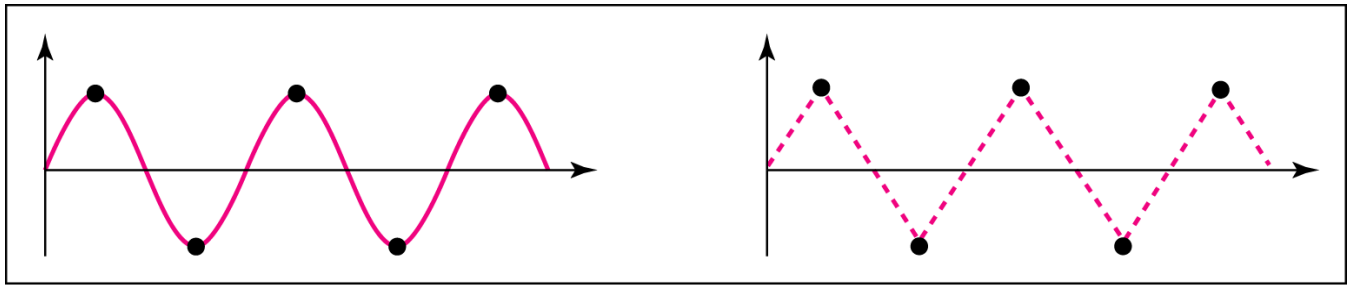
Sampling of Analog Signals

- It is impossible to send every single details of analog signals as digital. Only a sample of the signals should be sufficient to be sent over a channel. Nyquist sampling rate: the sampling rate must be **at least 2 times the highest frequency**
- The sampling of analog signals is also called 'Pulse Amplitude Modulation (PAM)'

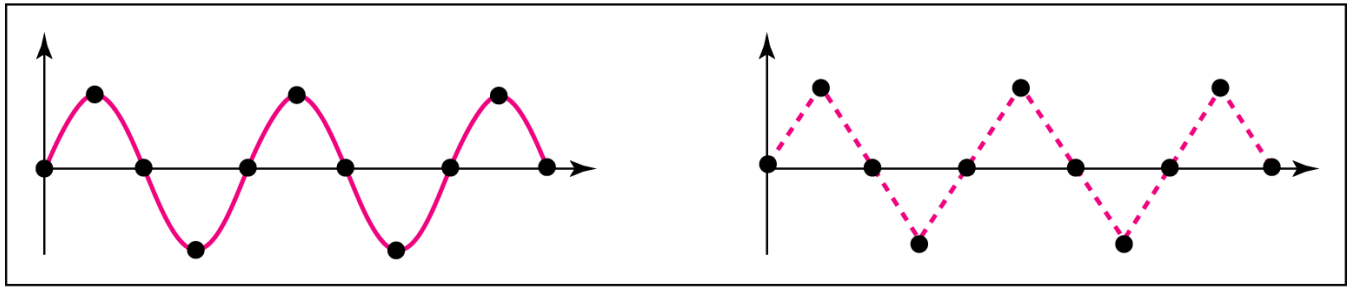


Each measurement is assigned a number (byte) according to its amplitude. The end result is a file comprising a string of bytes, eg ...
1001 1110 0001 1010 0111 0100 1111 1101 etc

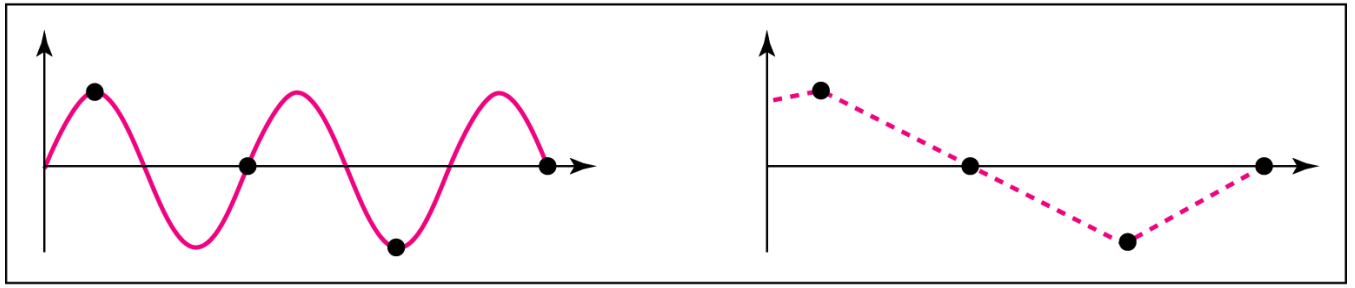
Recovery Signals at Different Sampling Rates



a. Nyquist rate sampling: $f_s = 2\ f$



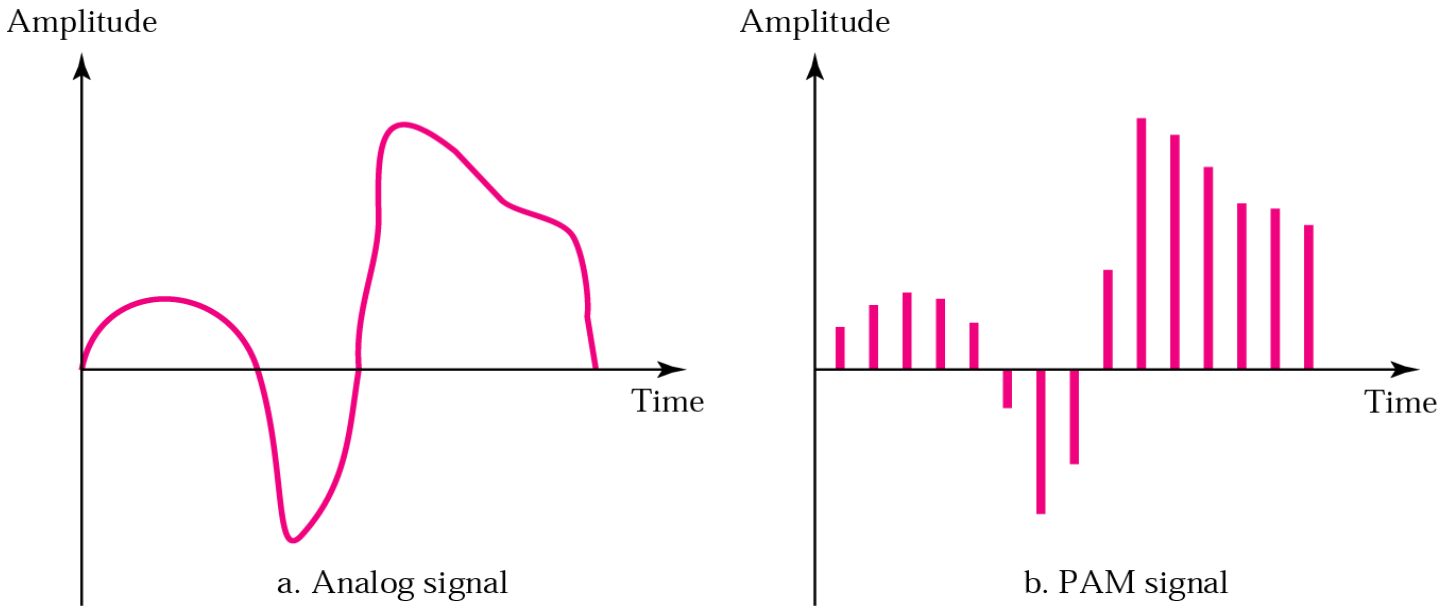
b. Oversampling: $f_s = 4\ f$



c. Undersampling: $f_s = f$

Pulse Amplitude Modulation

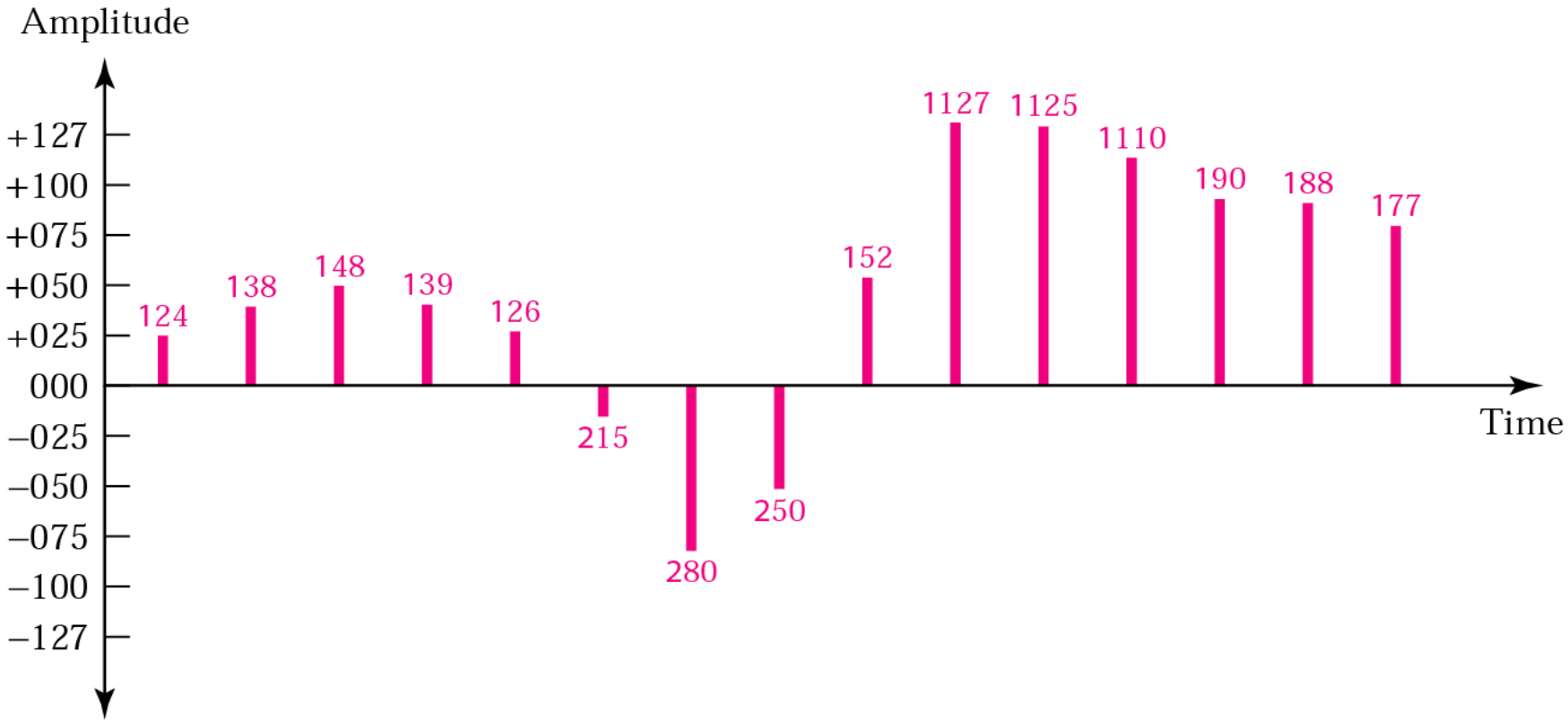
- Note: Pulse Amplitude Modulation (PAM) generates a series of pulses with different amplitude. It does not digitize the amplitude of the original analog signal.



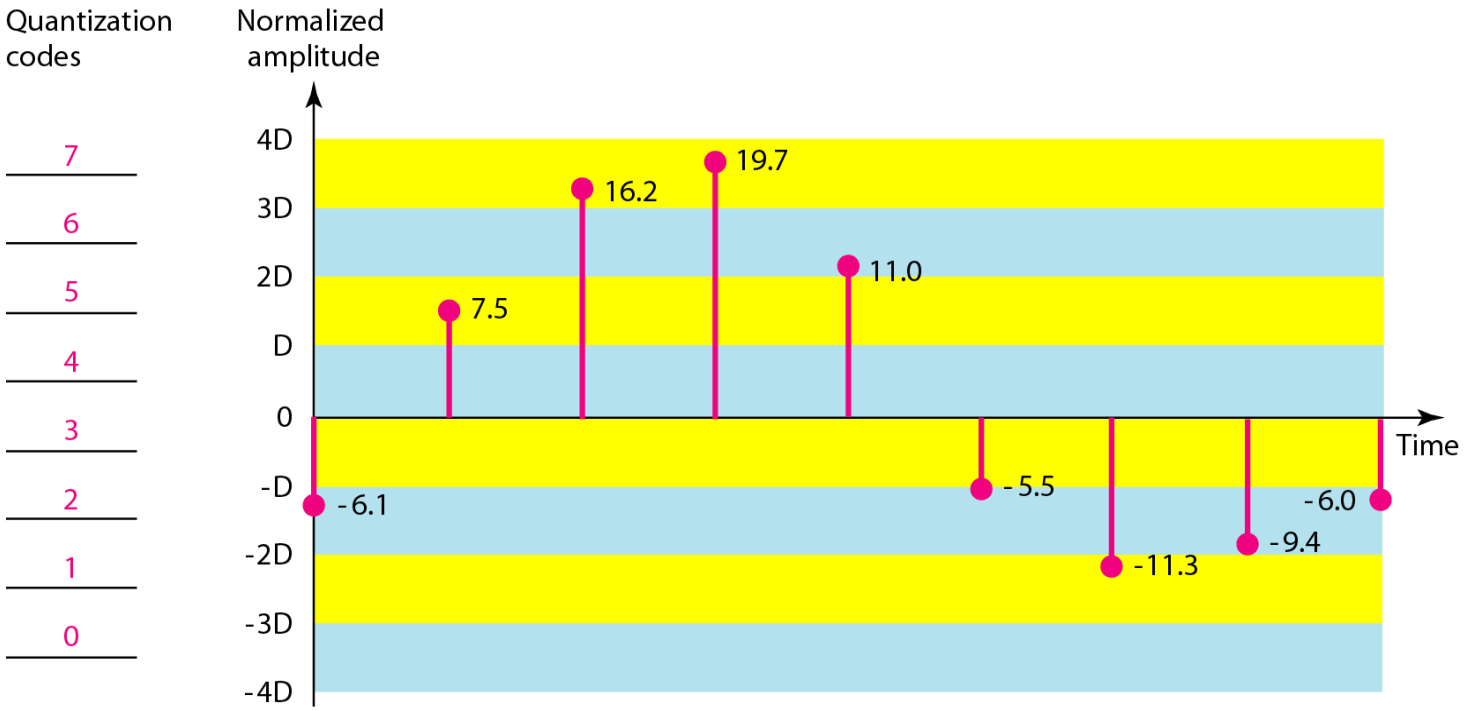
- Note: Pulse Amplitude Modulation (PAM) is performed prior to signal transmission.

Quantization

- We can **digitize the pulses into base-2 levels** so that we can send each pulse as bits. This process is called **quantization**.

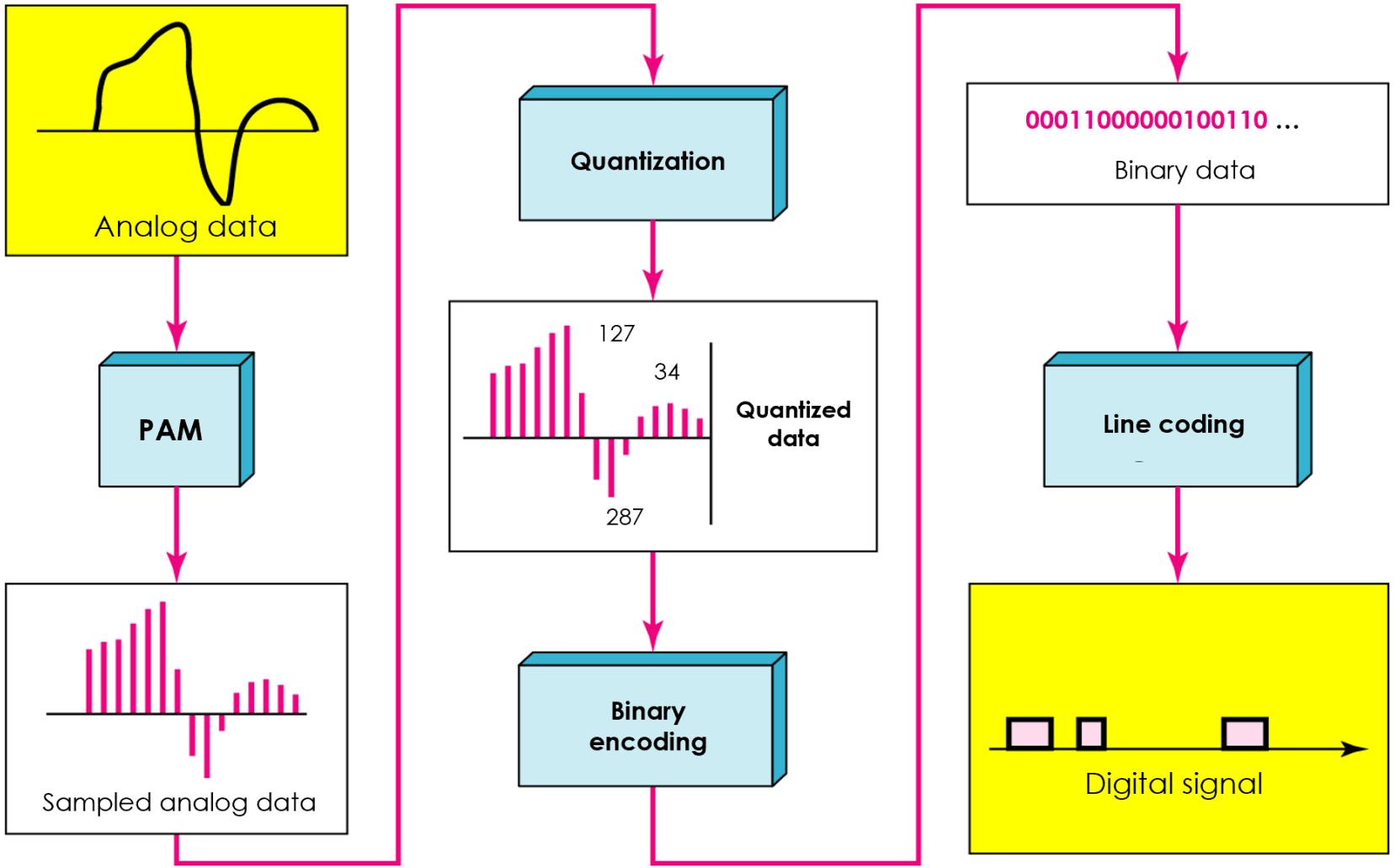


Pulse Code Modulation



Normalized PAM values	-1.22	1.50	3.24	3.94	2.20	-1.10	-2.26	-1.88	-1.20
Normalized quantized values	-1.50	1.50	3.50	3.50	2.50	-1.50	-2.50	-1.50	-1.50
Normalized error	-0.38	0	+0.26	-0.44	+0.30	-0.40	-0.24	+0.38	-0.30
Quantization code	2	5	7	7	6	2	1	2	2
Encoded words	010	101	111	111	110	010	001	010	010

Analog-to-Digital Conversion & Transmission



PAM, QUANTIZATION, PCM – SUMMARY

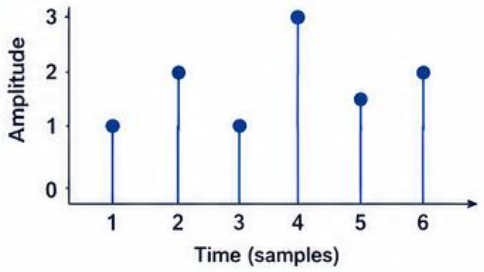
ICT

How an Analog Signal Becomes Digital Bits

1. PAM (Pulse Amplitude Modulation)

Information is represented by changing the **AMPLITUDE** (height) of pulses at each sampling instant.

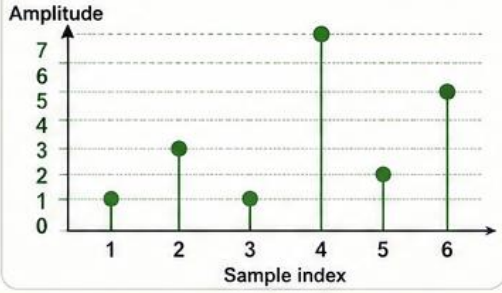
PAM Waveform (samples)



2. QUANTIZATION

Each sampled amplitude is mapped to the nearest value from a finite set of discrete levels.

Quantization Levels (example: 3 bits)



3. PCM (Pulse Code Modulation)

Quantized levels are encoded into binary codes to produce a digital bit stream.

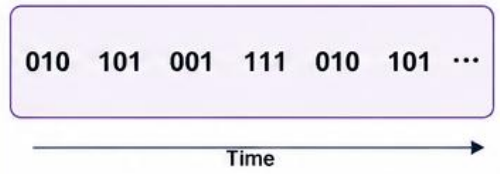
Encoding (example: 3 bits)

Level		3-bit Binary Code
7	→	111
6	→	110
5	→	101
4	→	100
3	→	011
2	→	010
1	→	001
0	→	000

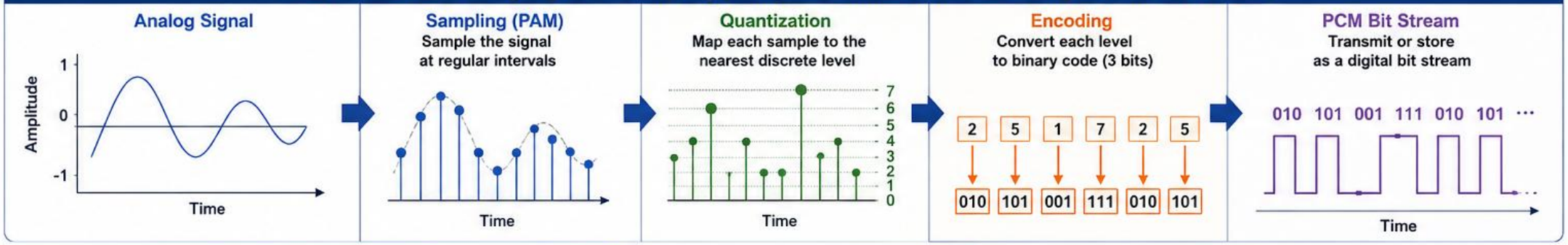
4. RESULT: DIGITAL BIT STREAM

The sequence of binary codes represents the original analog signal in digital form.

PCM Bit Stream (3 bits/sample)



THE PROCESS: Analog Signal → Sampling (PAM) → Quantization → Encoding → PCM Bit Stream



EXAMPLE (3-bit Quantization)

Sample Index	1	2	3	4	5	6
Analog Value (sampled)	2.3	4.7	0.9	6.8	1.6	4.9
Quantized Level	2	5	1	7	2	5
3-bit Binary Code	010	101	001	111	010	101

- **PAM:** Represents samples by pulse amplitude.
- **Quantization:** Rounds each sample to the nearest discrete level.
- **PCM:** Converts those levels to binary codes.

KEY POINTS

- ☑ PAM changes the pulse AMPLITUDE.
- ☑ **Quantization** reduces infinite values to a finite set of levels.
- ☑ PCM converts those levels into binary numbers.
- ☑ More bits (levels) → Higher accuracy, but more data.

PAM is about pulse amplitude, Quantization is about rounding to discrete levels, and PCM is the resulting digital representation.

Example

- We want to digitize the human voice (frequency 0 to 4,000 Hz). What is the bit rate, assuming 8 bits per sample?

- The human voice normally contains frequencies from 0 to 4000 Hz.

Sampling rate = 4000 x 2 = 8000 samples/second

- One sample is 8 bit.
- 8000 samples/second is therefore 8000 x 8 = 64,000 bits/second.

Example (2)

- We want to digitize a signal (frequency 2,000 to 8,000 Hz). What is the bit rate, assuming 8 bits per sample?

Sampling rate = _____ x 2 = _____ samples/second

- One sample is 8 bit.
- _____ samples/second is therefore _____ x 8 = _____Kbits/second.

Summary

- Digital-to-Digital Conversion Issues
  - Baseline wandering
  - DC components
  - Self-synchronization
- Line Coding
  - NRZ, NRZ-I, RZ
  - Manchester Differential Manchester
  - AMI
  - Block Coding (4B/5B)
- Signal Rate vs Data Rate
- Analog-to-Digital Conversion
  - Sampling
  - PAM
  - Quantization
  - PCM

